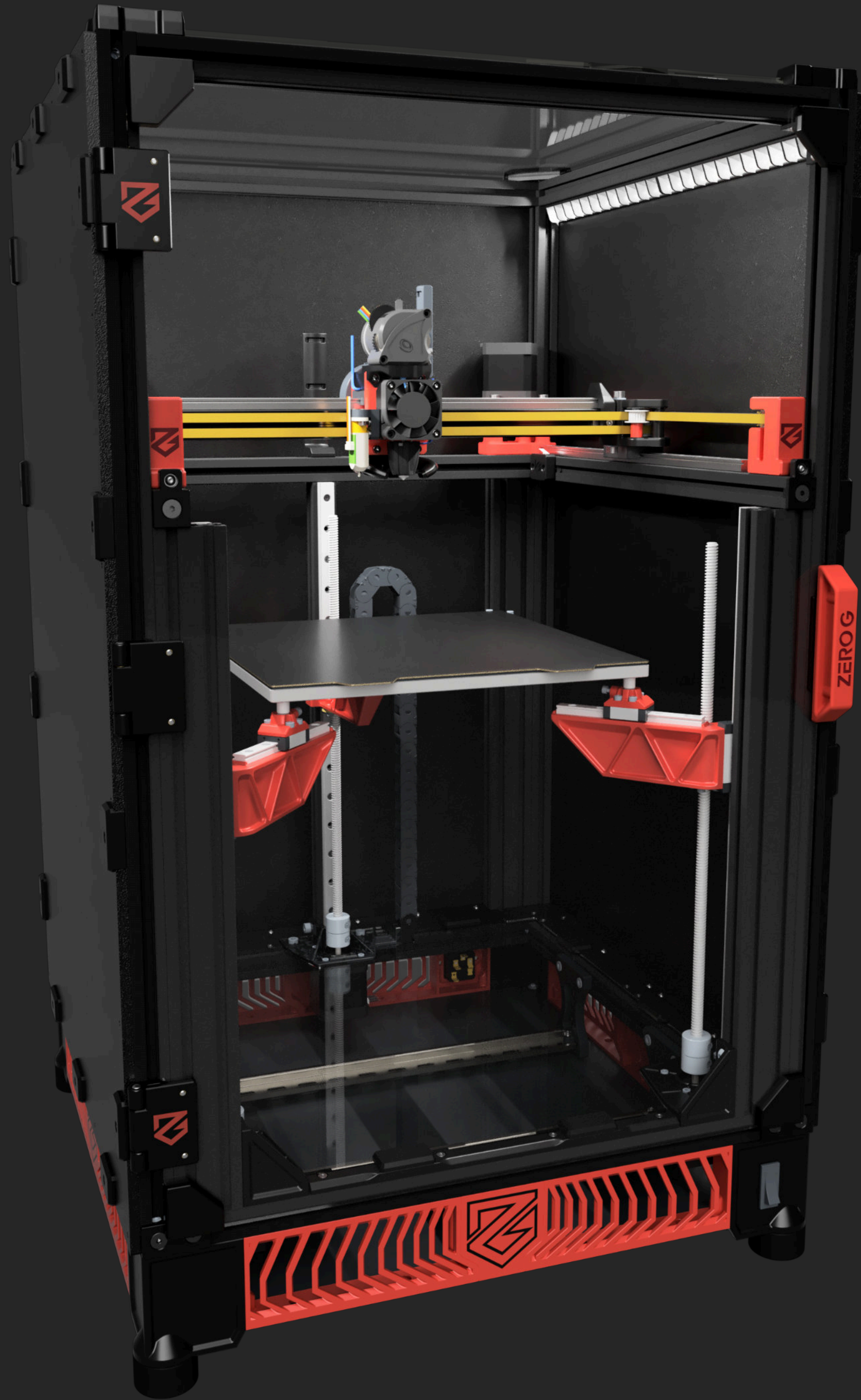


NEBULA

A Zero G Enclosure
For The Mercury One.1
and Hydra



NEBULA 255

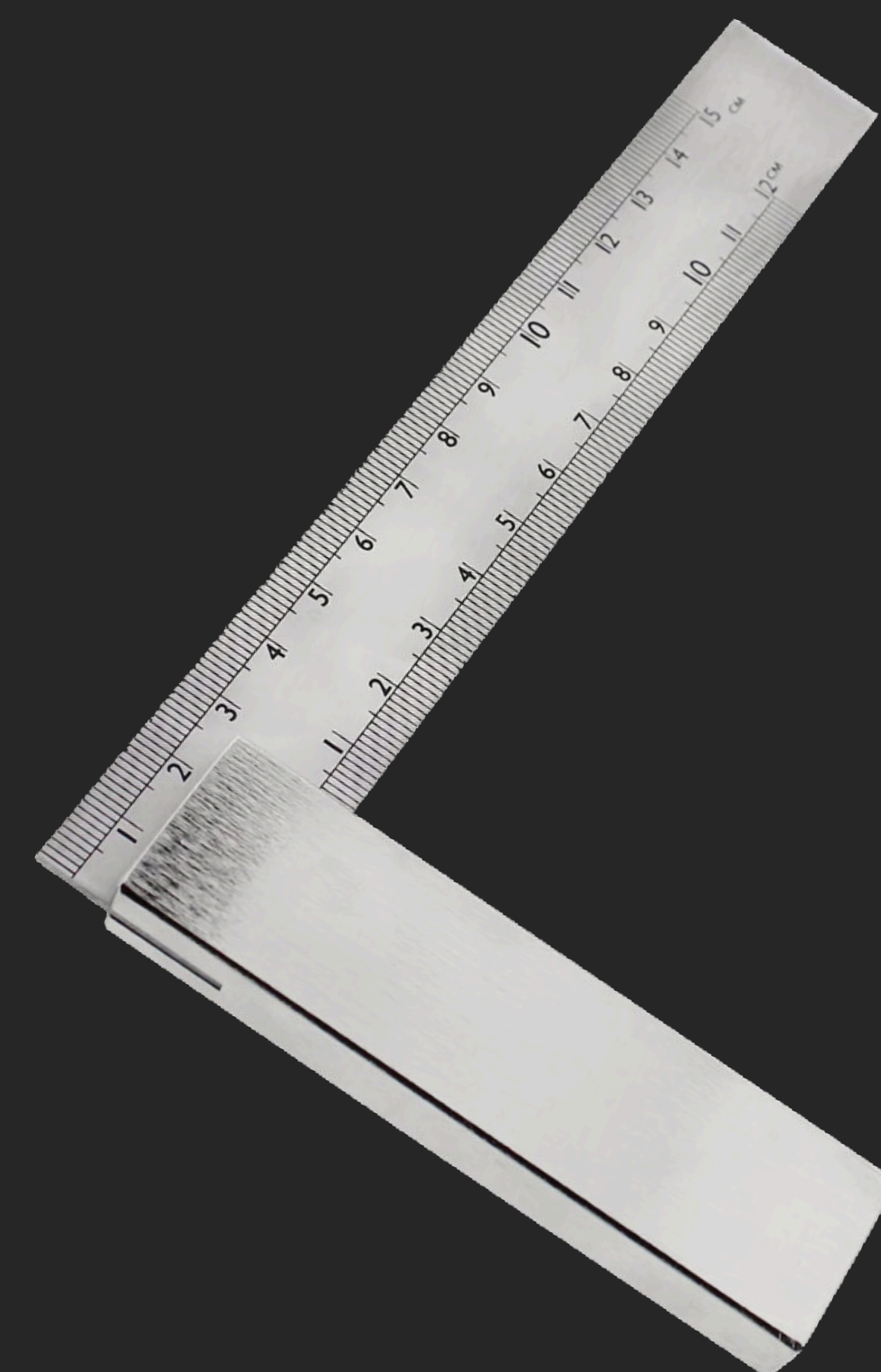


NEBULA 370

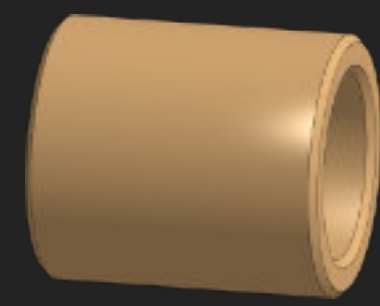
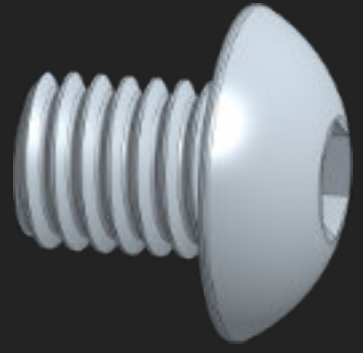
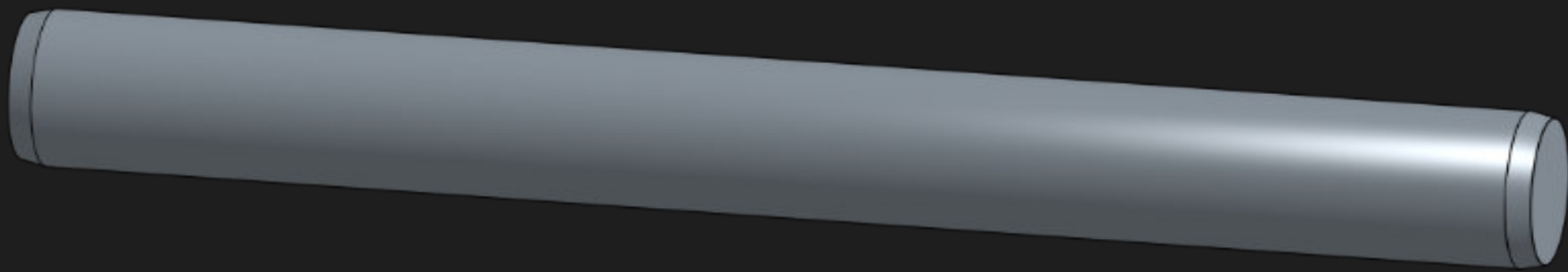
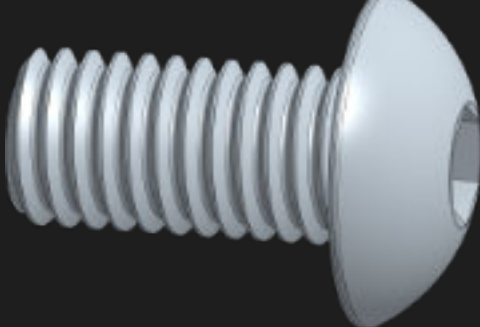
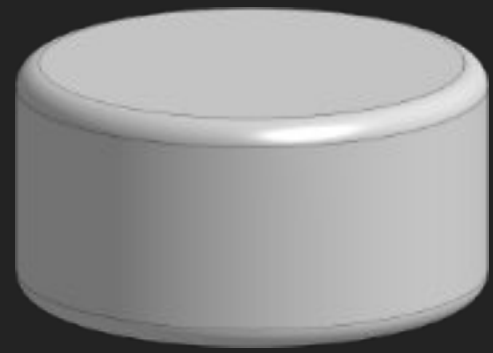
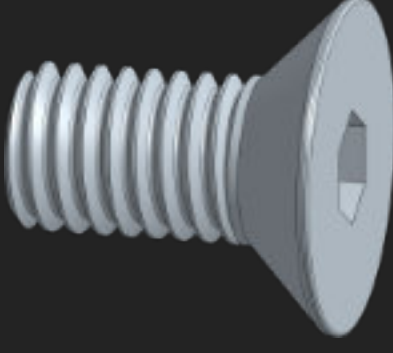
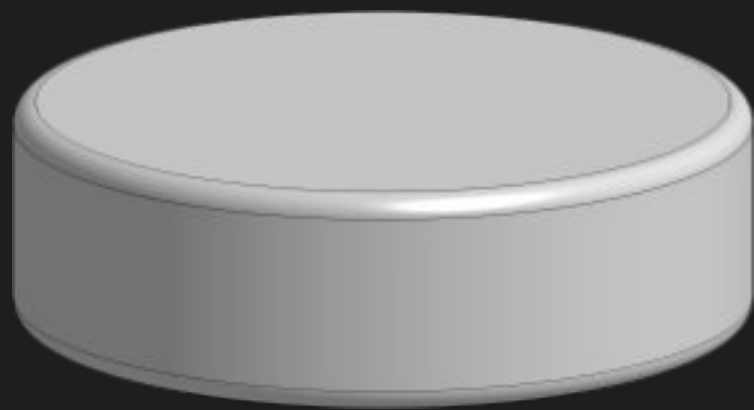
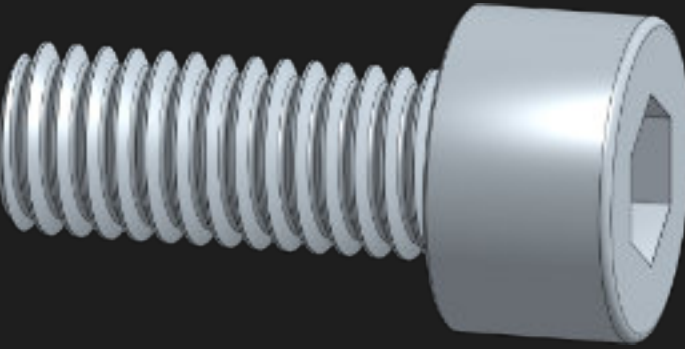
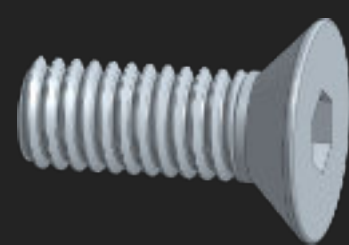
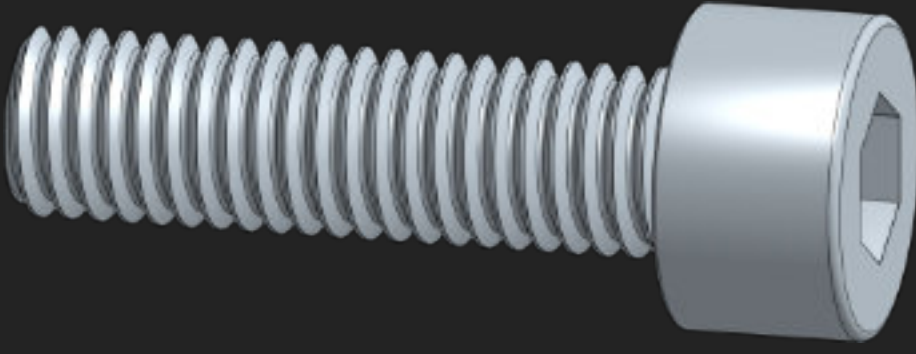
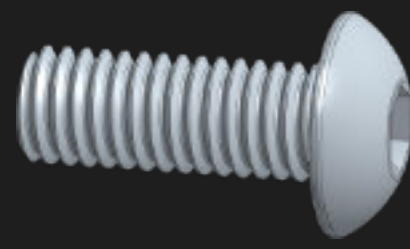
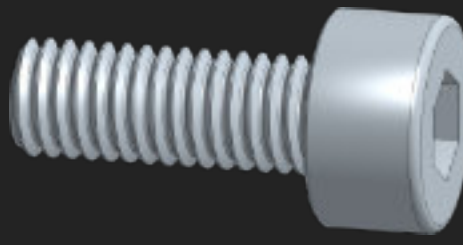
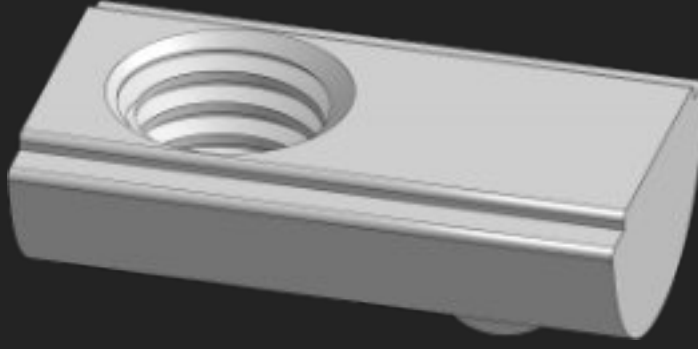
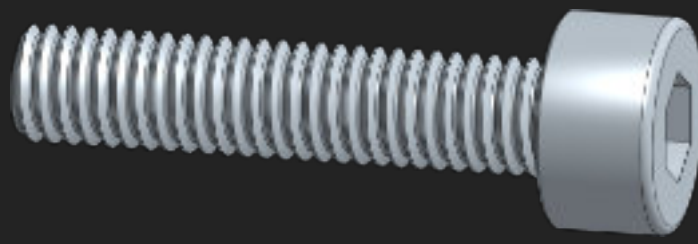
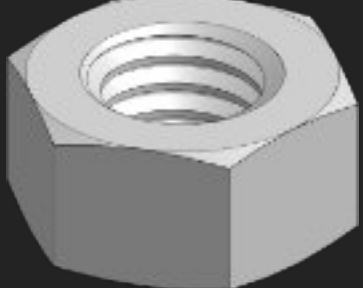
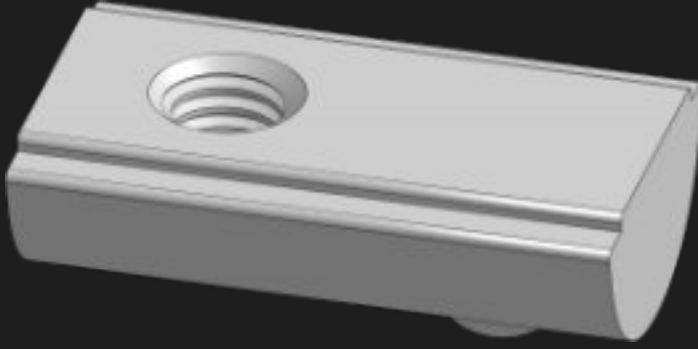


Included in this manual is the step-by-step guide and BOM for the Nebula 255 and 370. Both frames are assembled in the same way, with **hardware quantity** and **extrusion length differences**.

There are a few recommended tools that are required, and some additional items that are recommended.

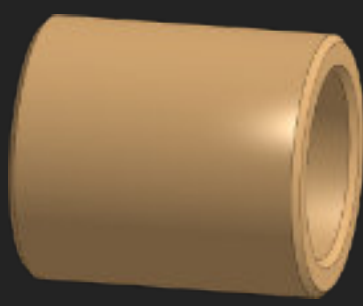
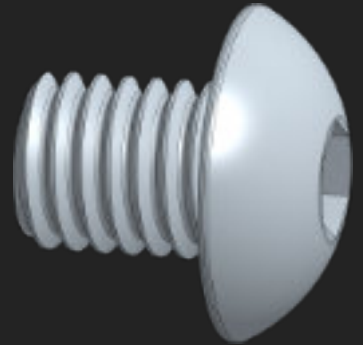
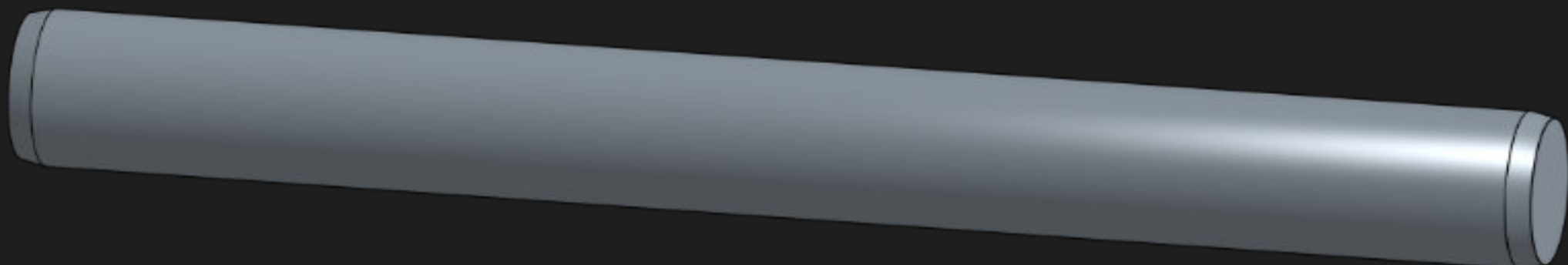
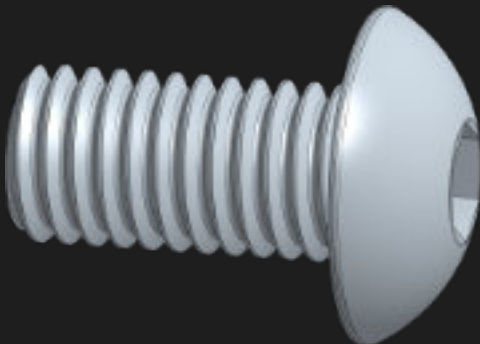
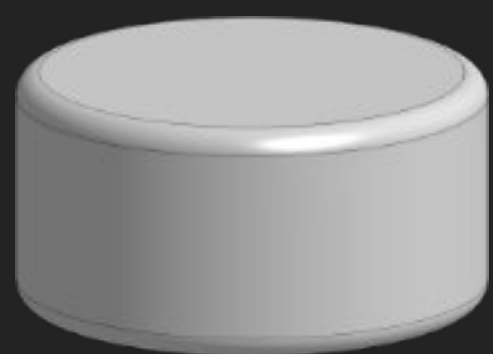
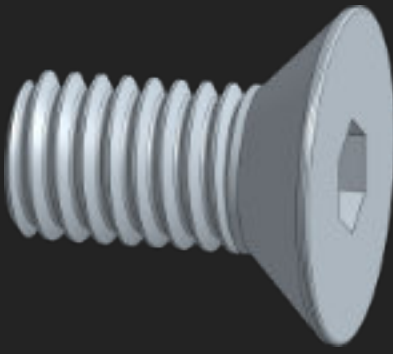
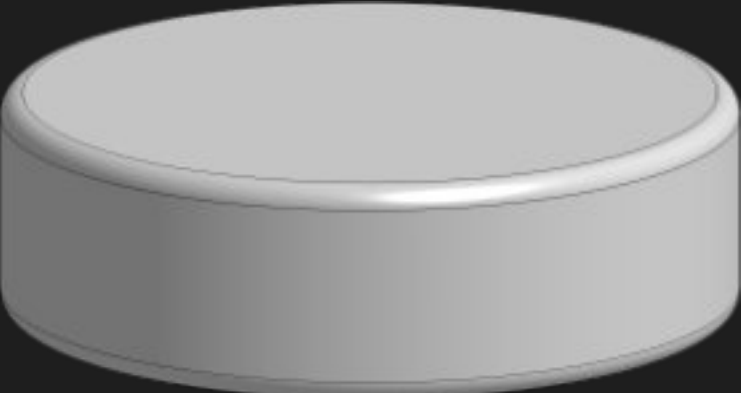
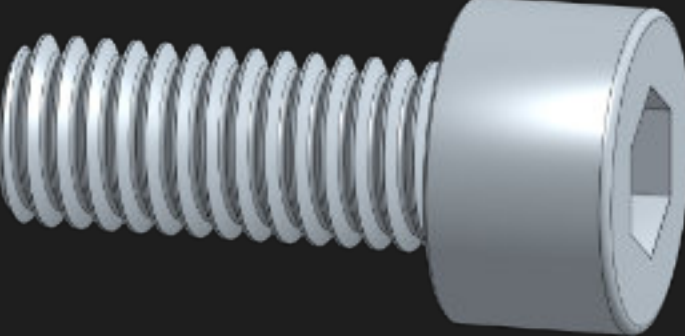
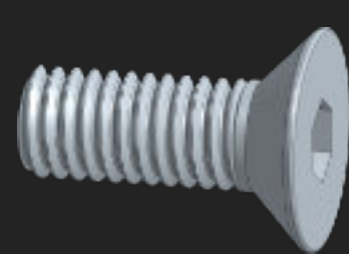
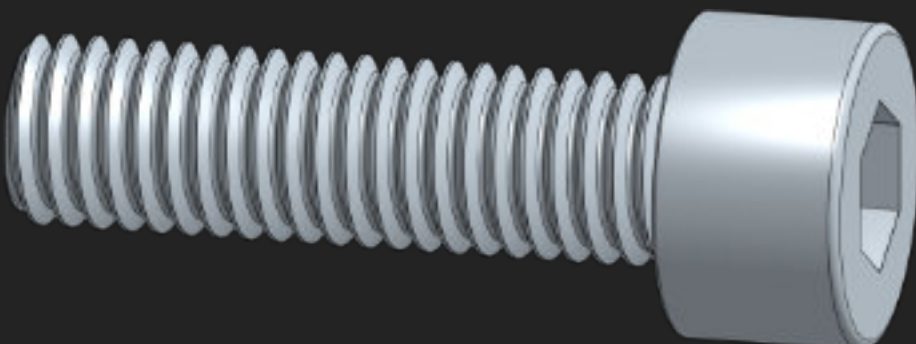
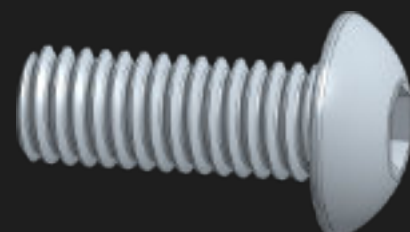
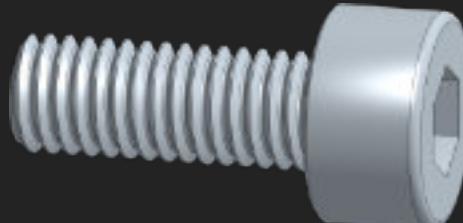
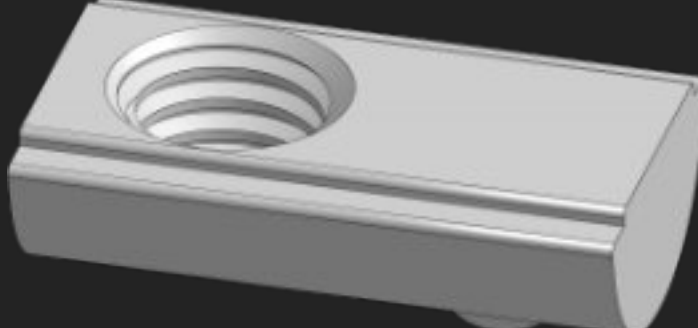
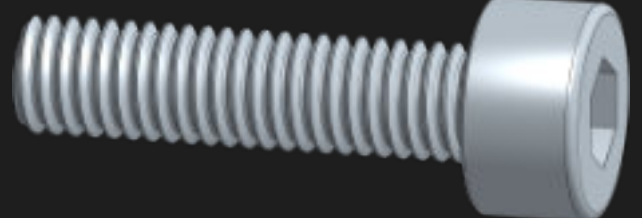
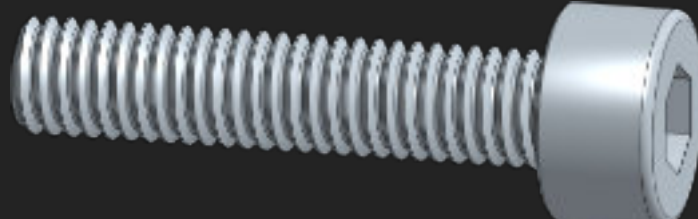
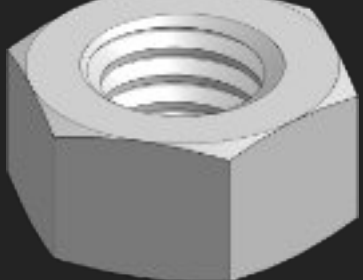
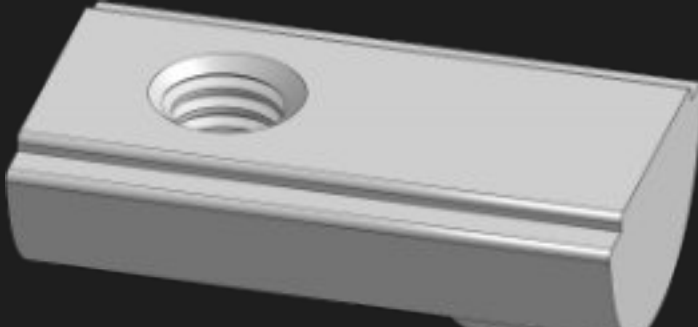


NEBULA 255 HARDWARE BOM

QTY	Part Description		QTY	Part Description	
6	5x7x8mm Bushing		8	M5x6 BHCS	
3	5x50 pin		66	M5x10 BHCS	
36	6x3 Magnet		6	M5x10 FHCS	
16	10x3 Magnet		4	M5x12 SHCS	
2	M3x8 FHCS		14	M5x18 SHCS	
36	M3x8 BHCS		26	M3 X D5.0 X L4.0 - Heat-set	
38	M3x8 SHCS		52	M5 Roll In T-Nut	
46	M3x12 SHCS		1	3x5mm Foam Tape	10 Meters
8	M3x14 SHCS		4	M5 Hex Nut	
106	M3 Roll In T-nut		4	38x19mm Rubber Foot	



NEBULA 370 HARDWARE BOM

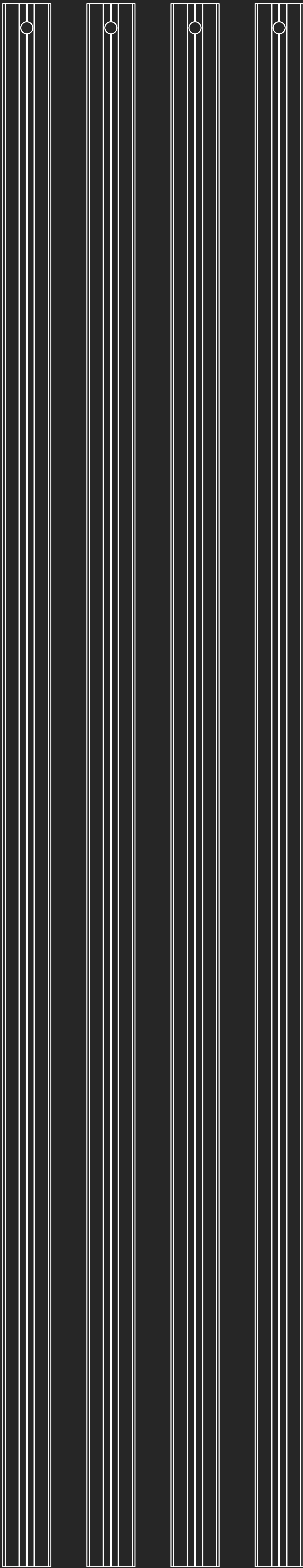
QTY	Part Description		QTY	Part Description	
12	5x7x8mm Bushing		8	M5x6 BHCS	
6	5x50 pin		66	M5x10 BHCS	
36	6x3 Magnet		6	M5x10 FHCS	
48	10x3 Magnet		4	M5x12 SHCS	
2	M3x8 FHCS		14	M5x18 SHCS	
44	M3x8 BHCS		34	M3 X D5.0 X L4.0 - Heat-set	
46	M3x8 SHCS		52	M5 Roll In T-Nut	
52	M3x12 SHCS		1	3x5mm Foam Tape	15 Meters
8	M3x14 SHCS		4	M5 Hex Nut	
118	M3 Roll In T-nut		4	38x19mm Rubber Foot	



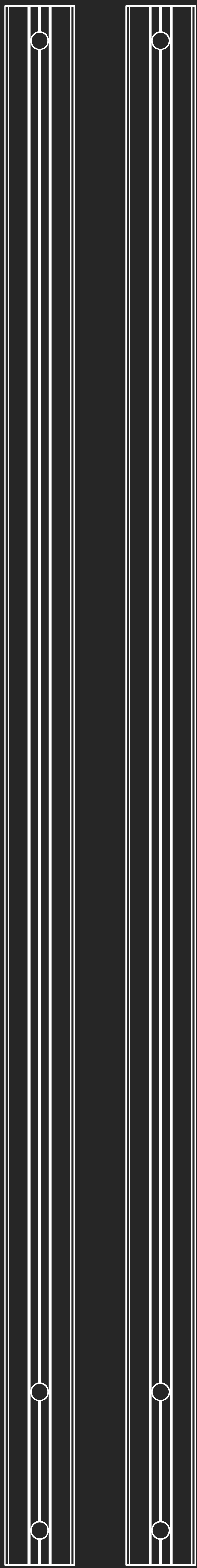
EXTRUSION BOM

Note: Extrusions D and E are the same length for the Nebula 255, but not the same length for the Nebula 370.

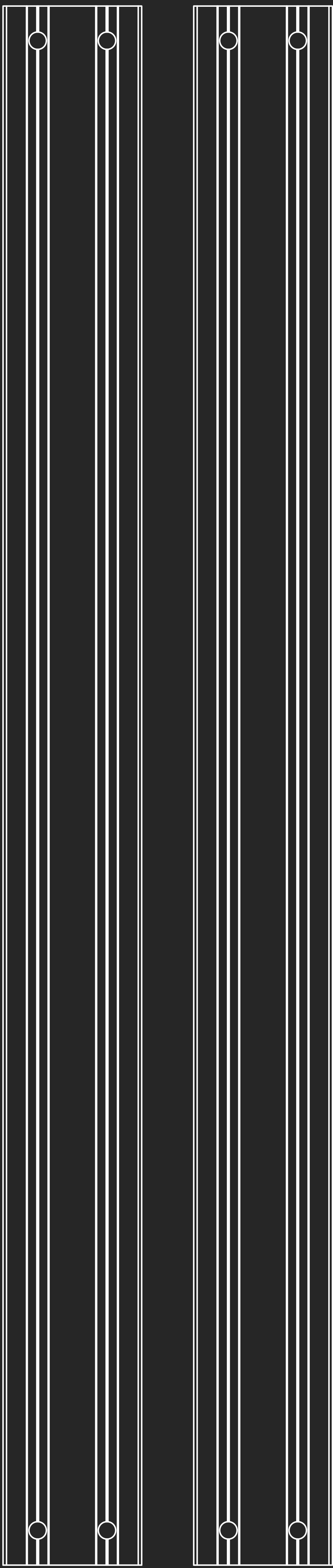
	QTY.	Nebula 255	Nebula 370
A	4x	2020 650mm	2020 800mm
B	2x	2020 450mm	2020 600mm
C	2x	2040 450mm	2040 600mm
D	2x	2020 410mm	2020 560mm
E	2x	2020 410mm	2020 538mm
F	4x	2040 400mm	2040 510mm
G	1x	2020 400mm	2020 510mm
H	2x	2020 370mm	2020 498mm
I	3x	2020 370mm	2020 498mm



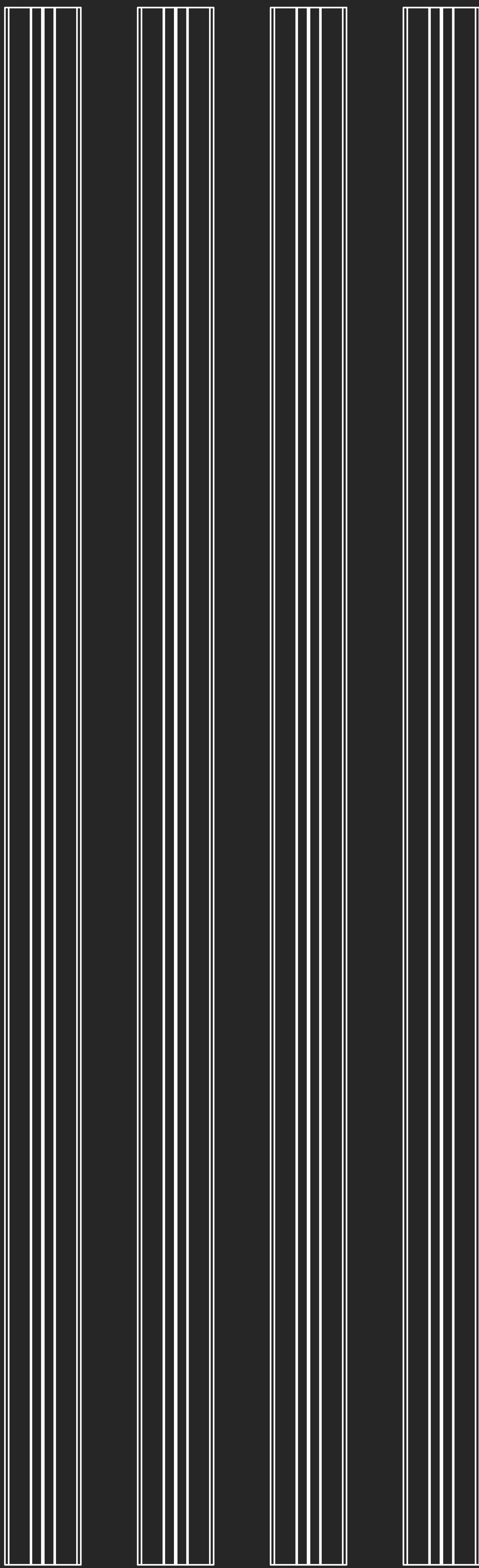
A



B

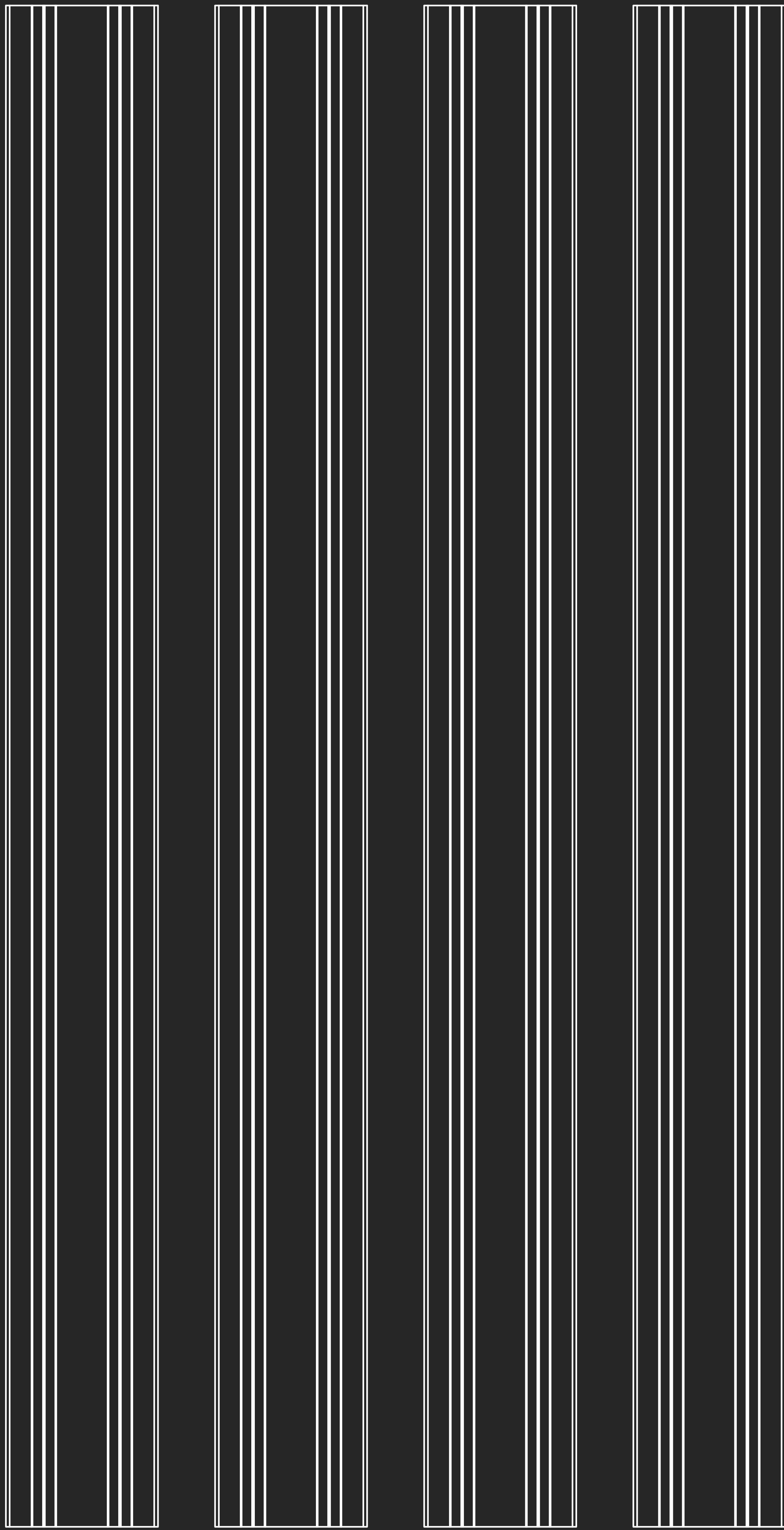


C

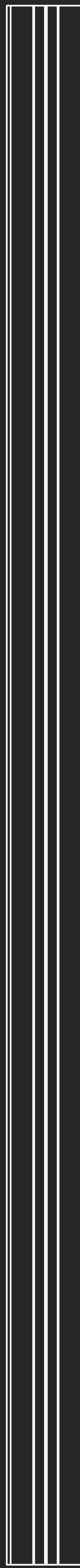


D

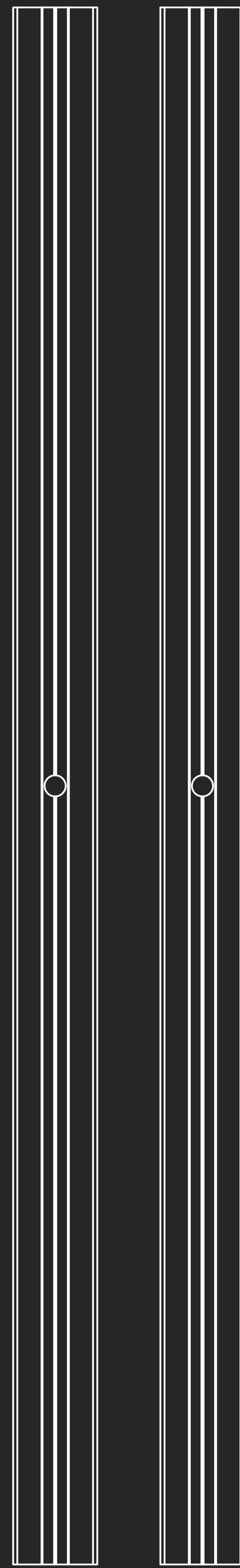
E



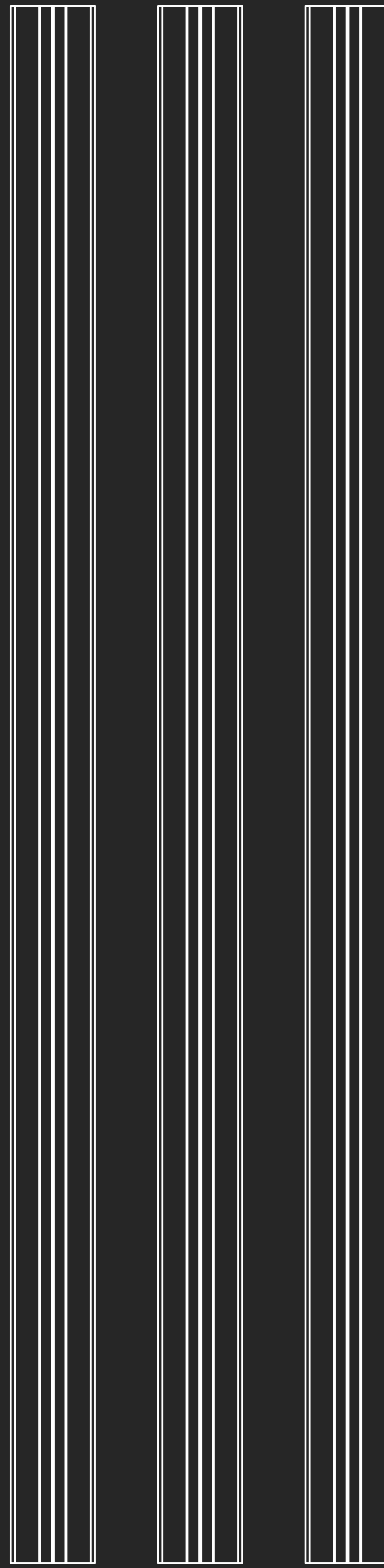
F



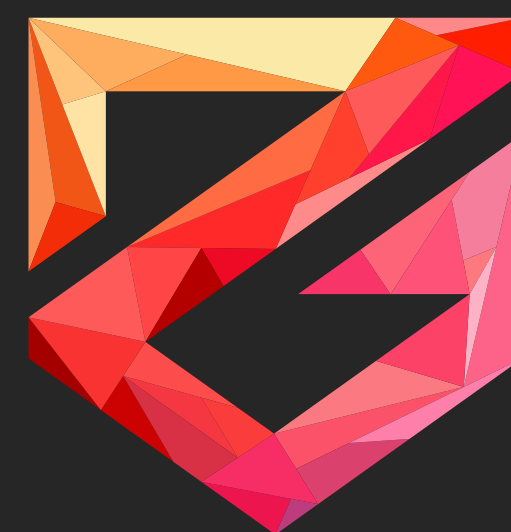
G



H

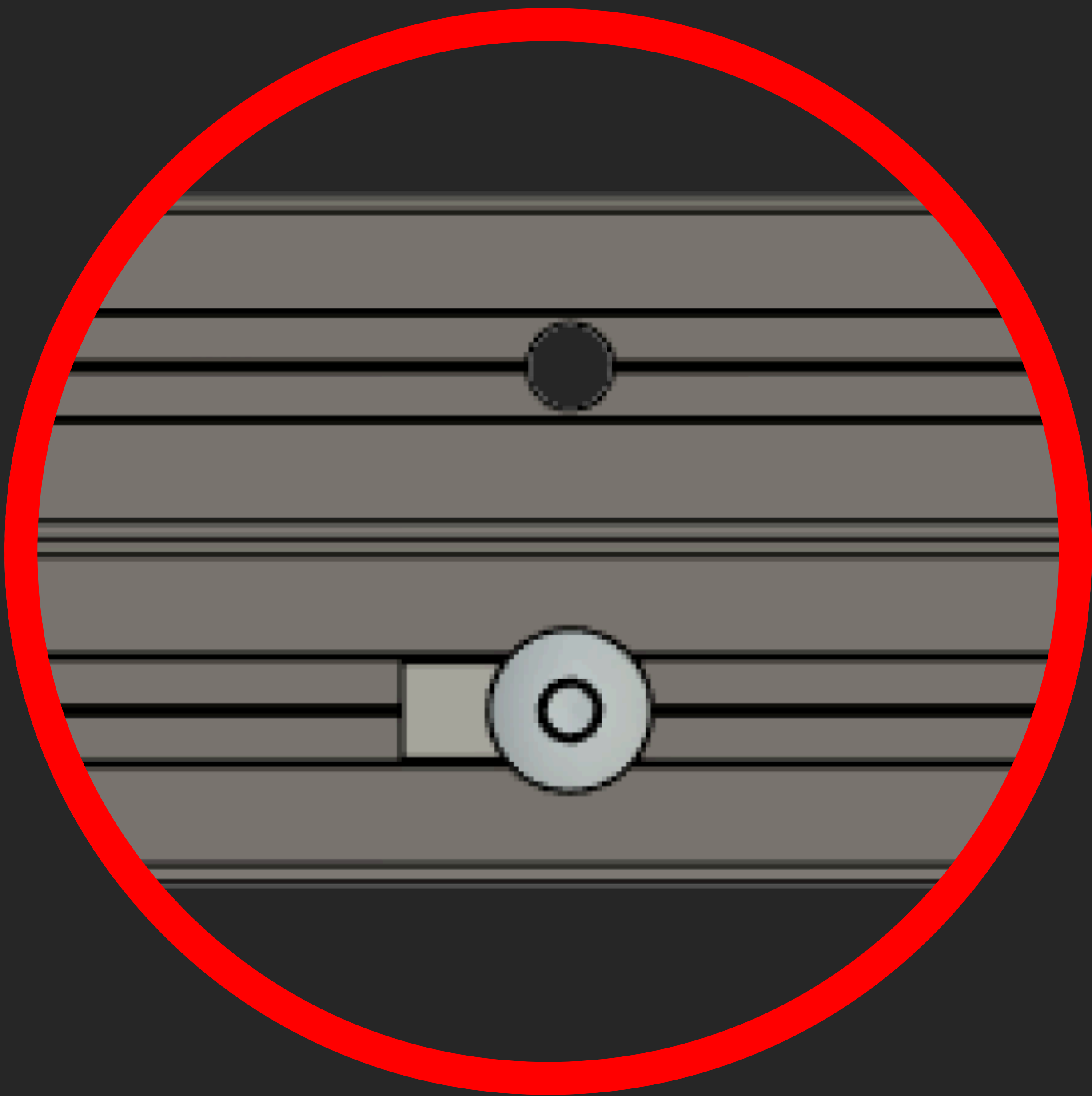


I



ASSEMBLING THE UPRIGHTS

	QTY.	Nebula 255	Nebula 370
A	4x	2020 650mm	2020 800mm
G	4x	2020 400mm	2020 510mm
	10x	M5 T-Nut	M5 T-Nut
	10x	M5x6 BHCS	M5x6 BHCS



- Place the 2020 upright next to the 2040 upright to position the 2 M5 T-nuts and 2 M5x6 BHCS.
- Do this for all 4 uprights.

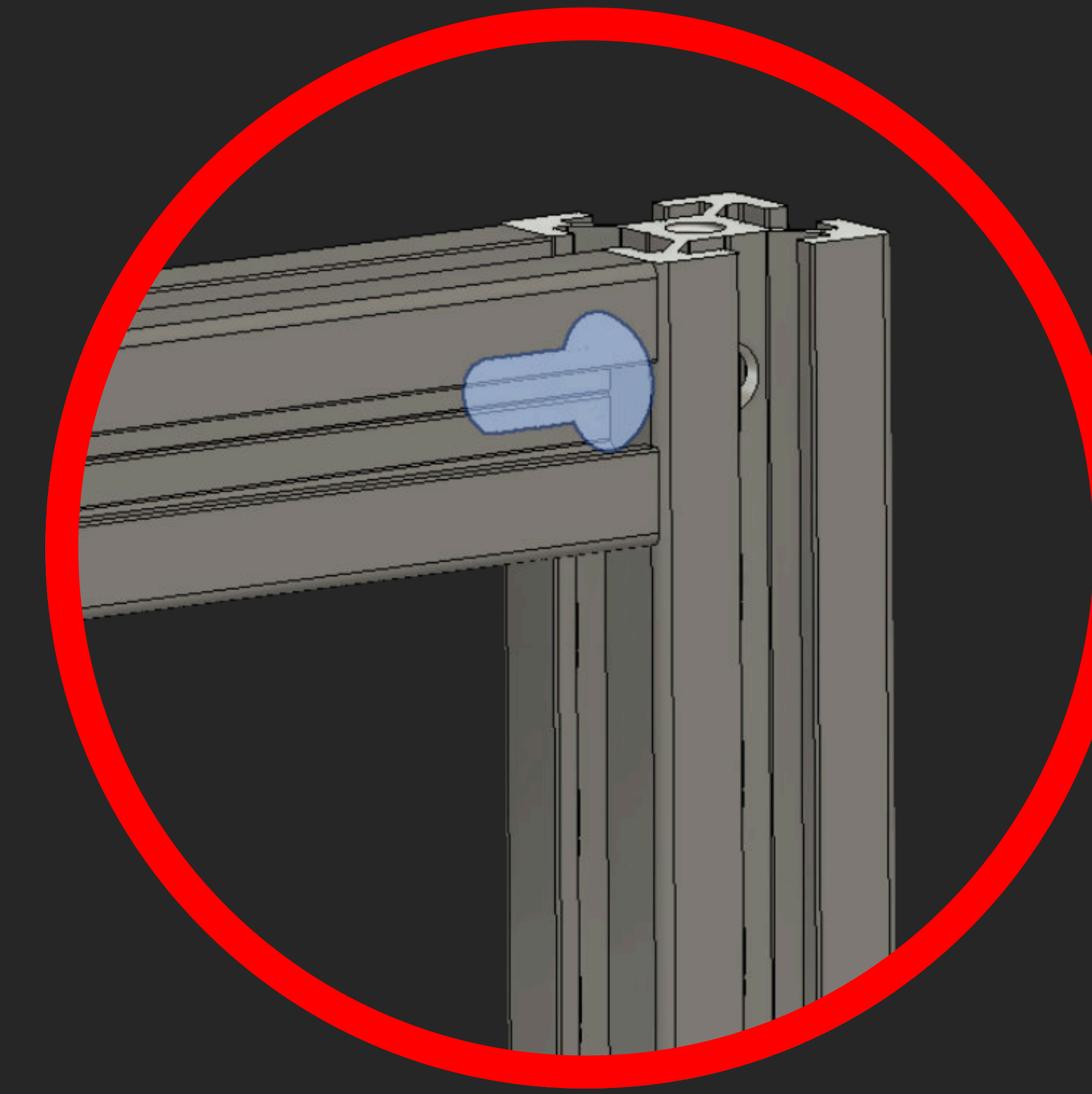
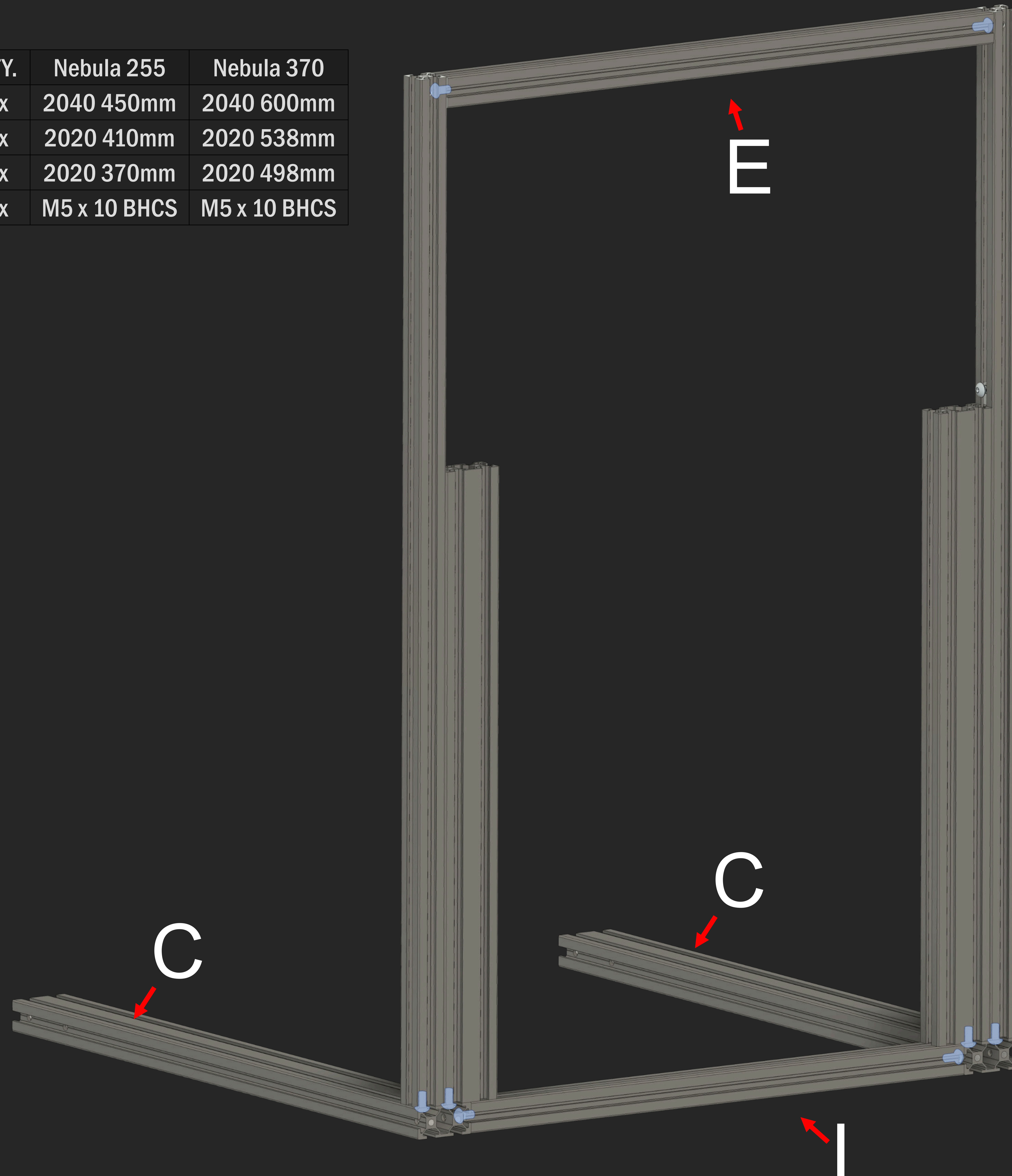


Slide in a M5 T-nut and M5x6 BHCS. This is for the Y Gantry extrusion



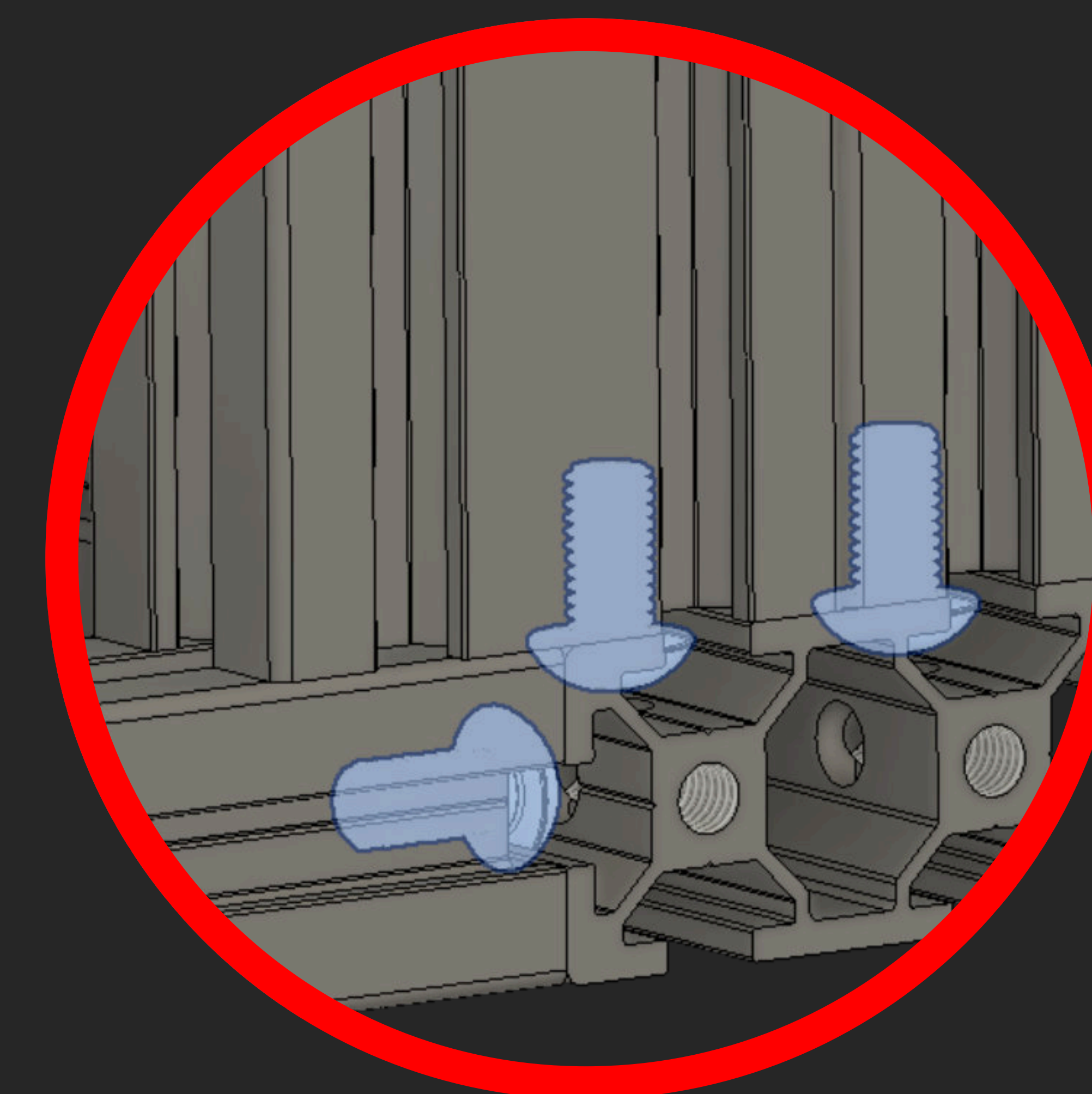
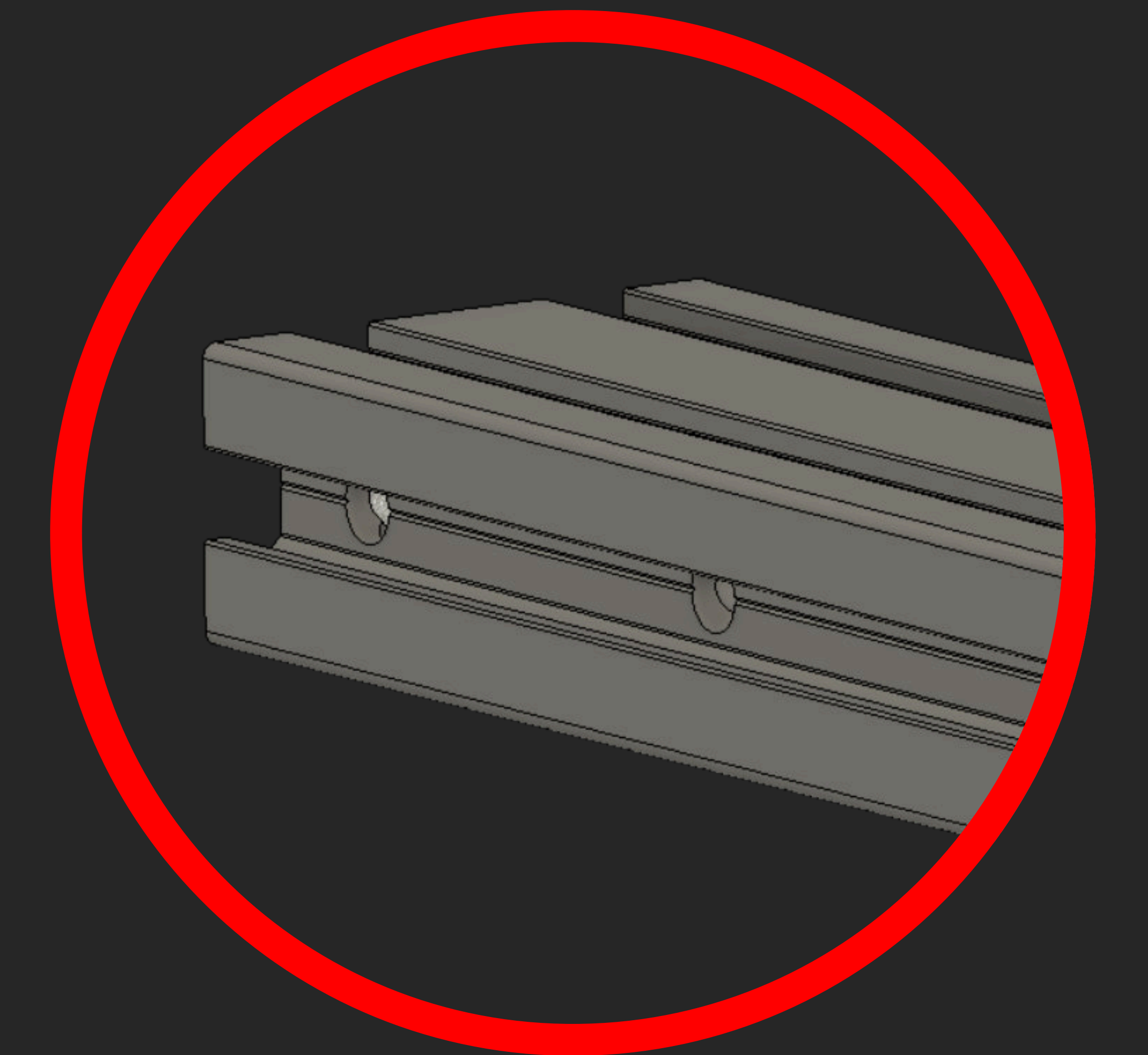
ASSEMBLING THE FRONT SECTION

	QTY.	Nebula 255	Nebula 370
C	2x	2040 450mm	2040 600mm
E	1x	2020 410mm	2020 538mm
I	1x	2020 370mm	2020 498mm
	8x	M5 x 10 BHCS	M5 x 10 BHCS



Start with the front section of the frame, install blind joints using M5 x 10 BHCS. The images on this page should help

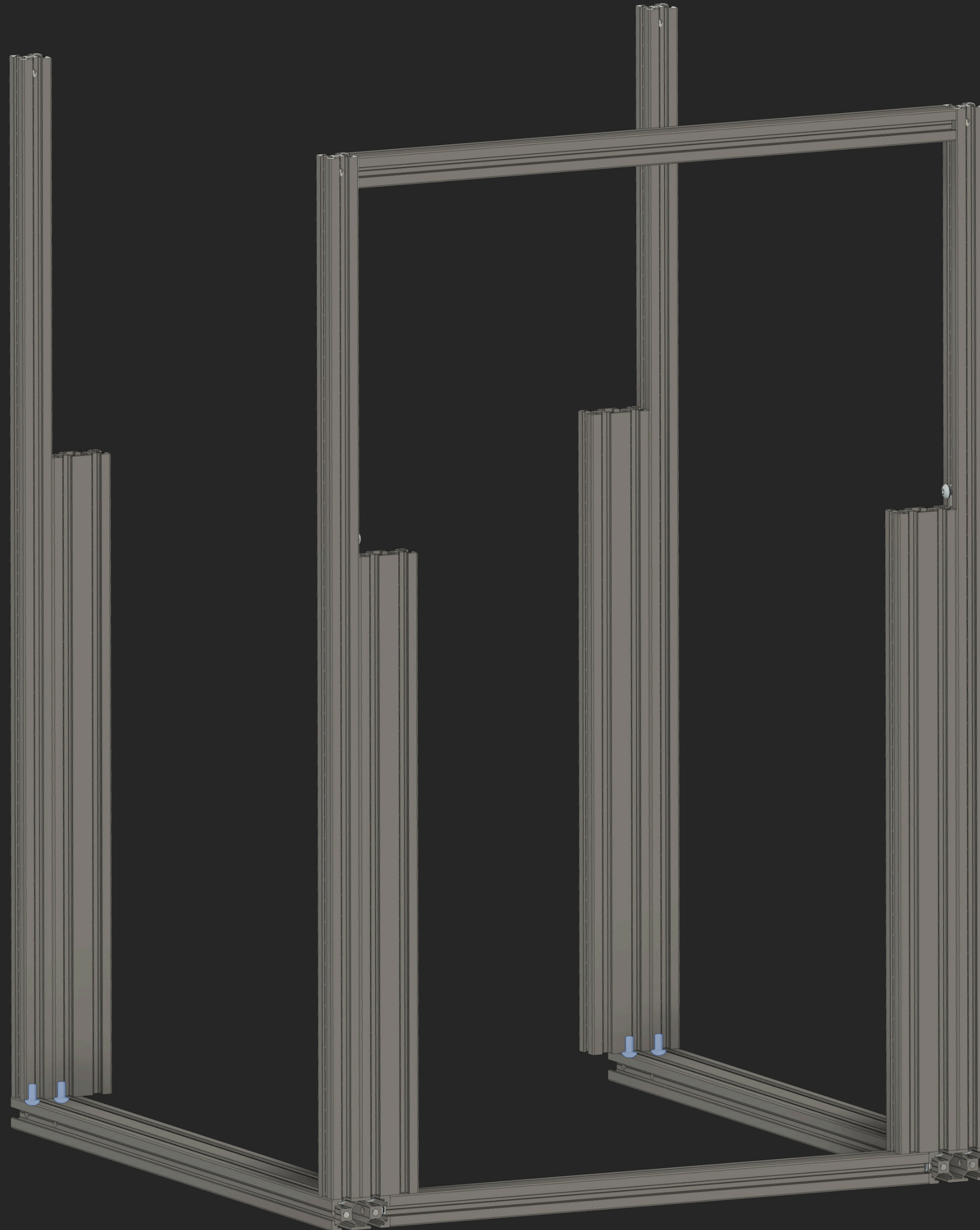
Make sure that the bottom 2040 extrusions have the end with the 2 blind cross holes (C Arrow position) in the back for Hydra.



Notice that the 2040 uprights only get one blind joint. Do not add a second.



INSTALL THE REAR UPRIGHTS



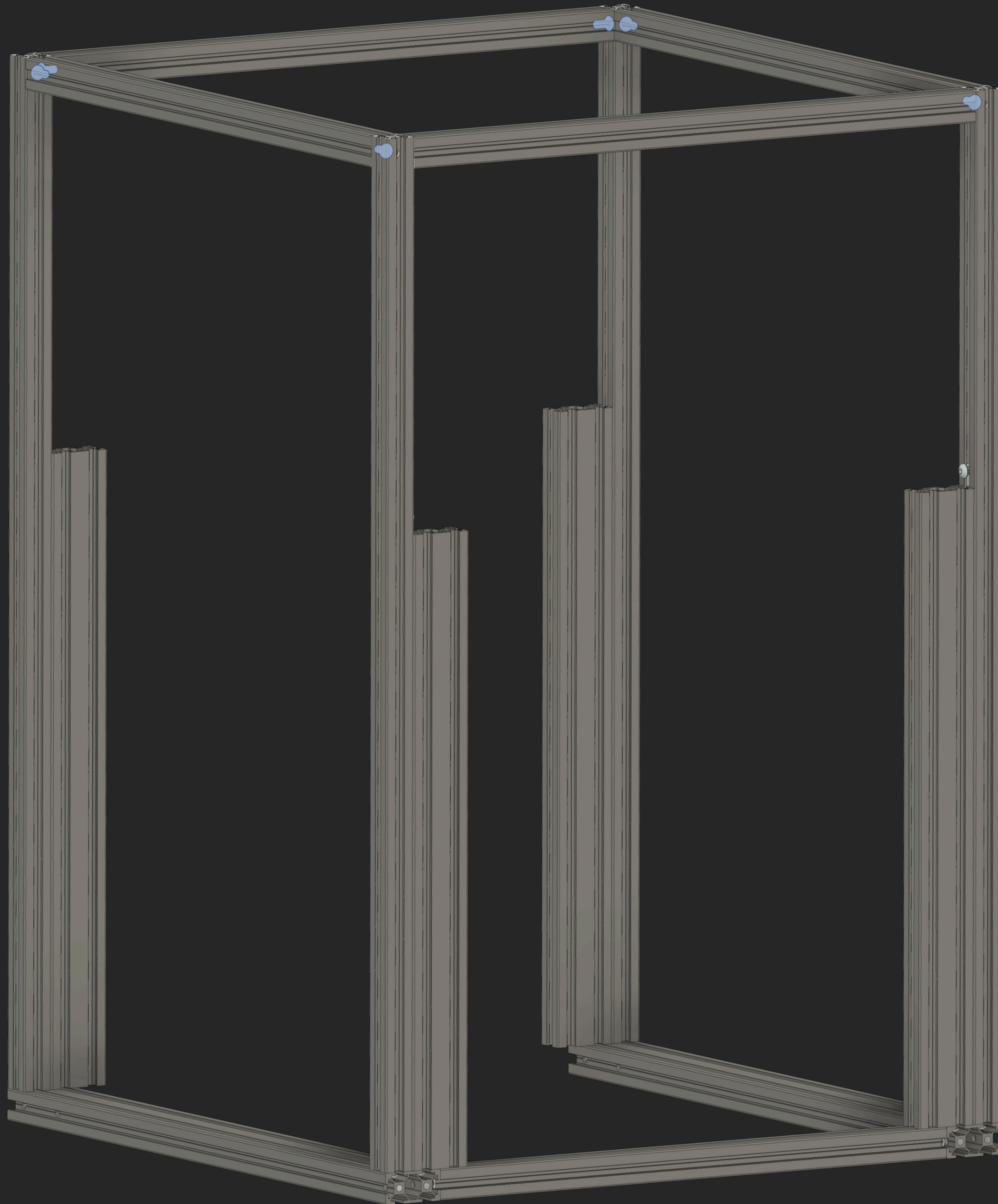
	QTY.	Nebula 255	Nebula 370
	4x	M5 x 10 BHCS	M5 x 10 BHCS

Now you can install the rear 2 uprights. We will add more support in the next step.

NOTE It is helpful to lay the frame on its side to help tighten the blind joints.



INSTALL THE TOP BRACES

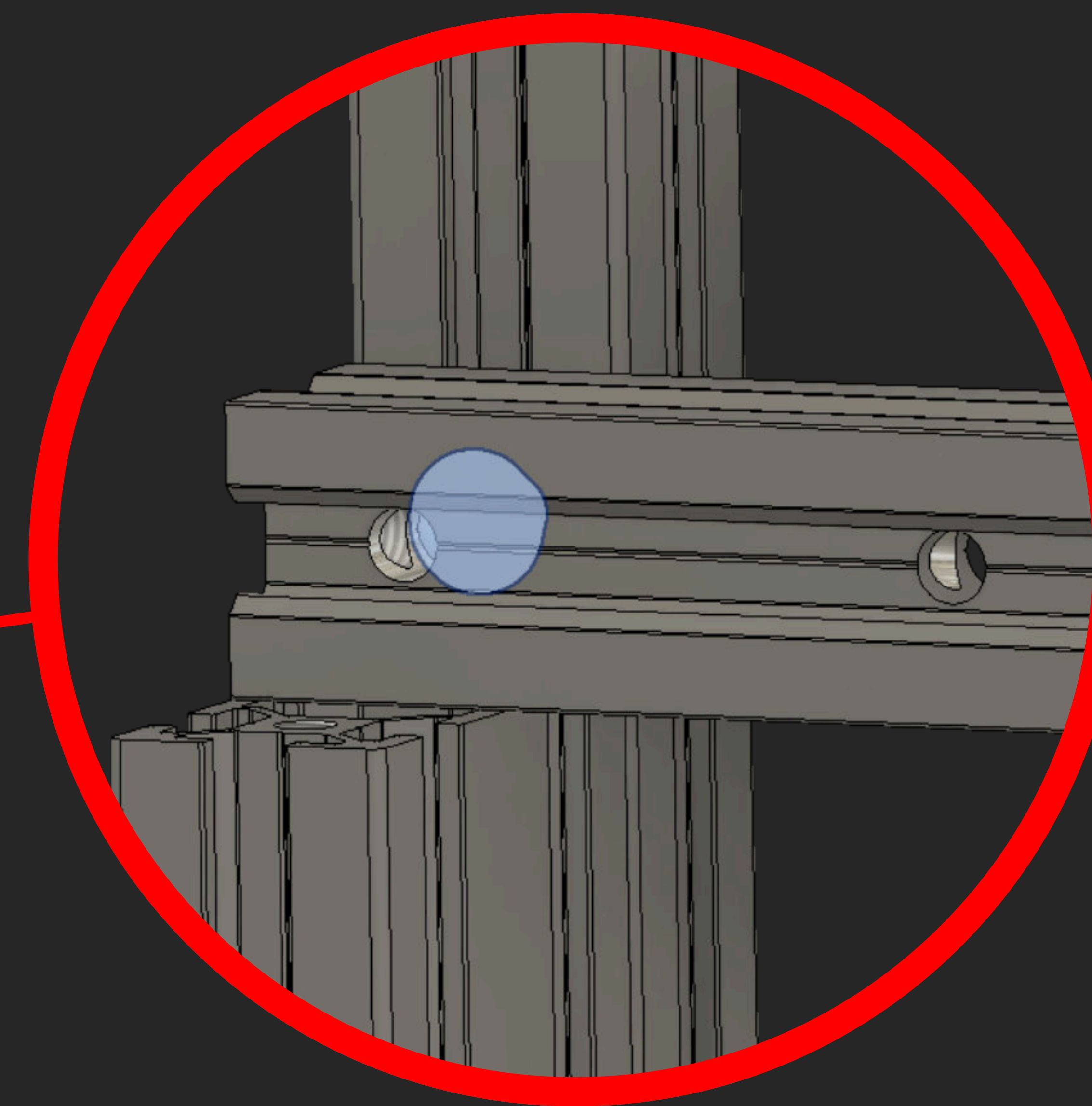
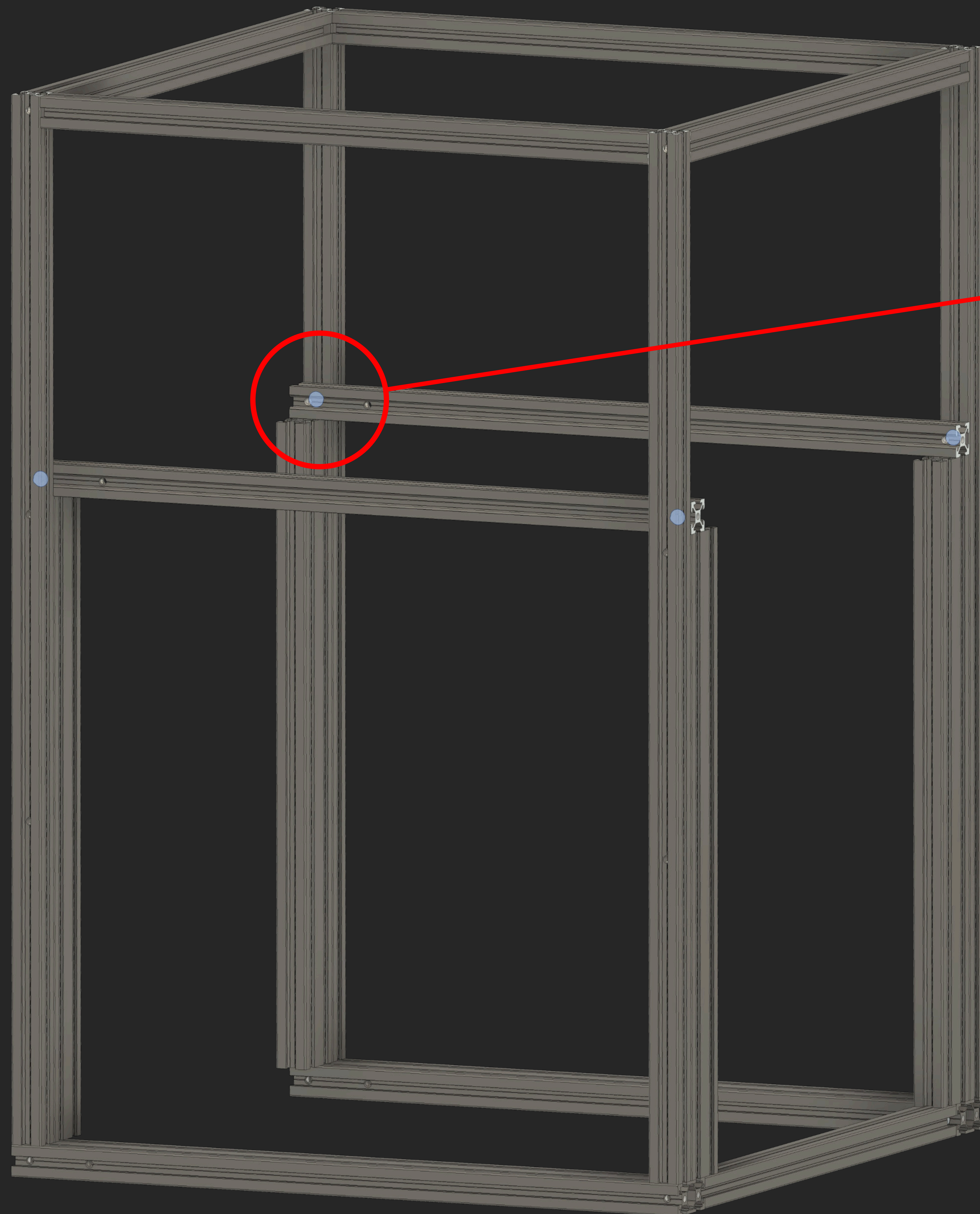


	QTY.	Nebula 255	Nebula 370
D	2x	2020 410mm	2020 560mm
E	1x	2020 410mm	2020 538mm
	6x	M5 x 10 BHCS	M5 x 10 BHCS

Add the remaining 3 top extrusions now using M5 x 10 BHCS.



INSTALLING Y GANTRY BARS



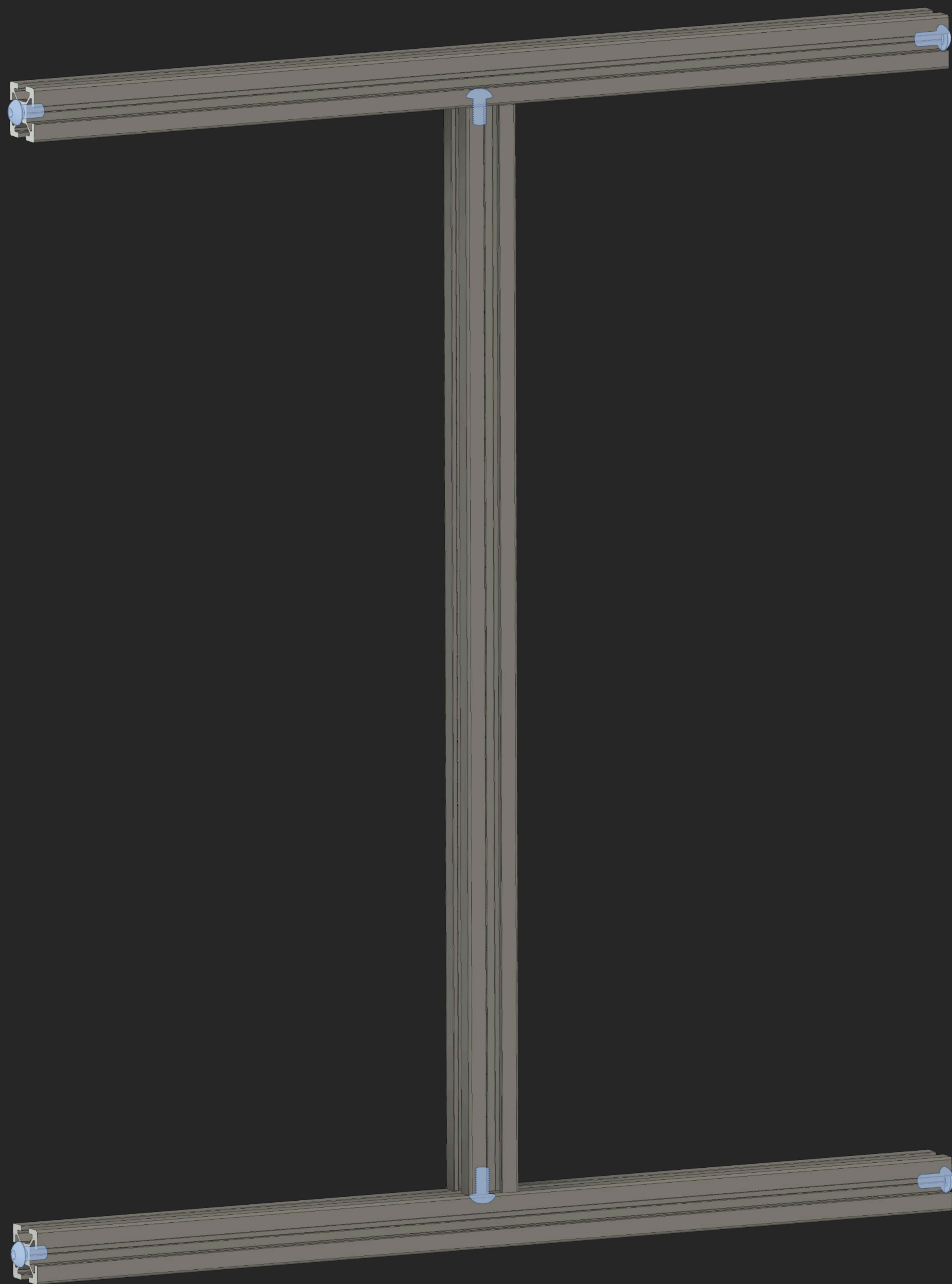
	QTY.	Nebula 255	Nebula 370
B	2x	2020 450mm	2020 600mm

Slide in the Y-Axis 2020 extrusions, using the T-Nut Blind Joints as a guide. Loosen or tighten the T-Nut joints as needed.

Make sure the two holes for the Hydra sub-frame are located toward the rear.



ASSEMBLE REAR HYDRA UPRIGHT SUB-FRAME



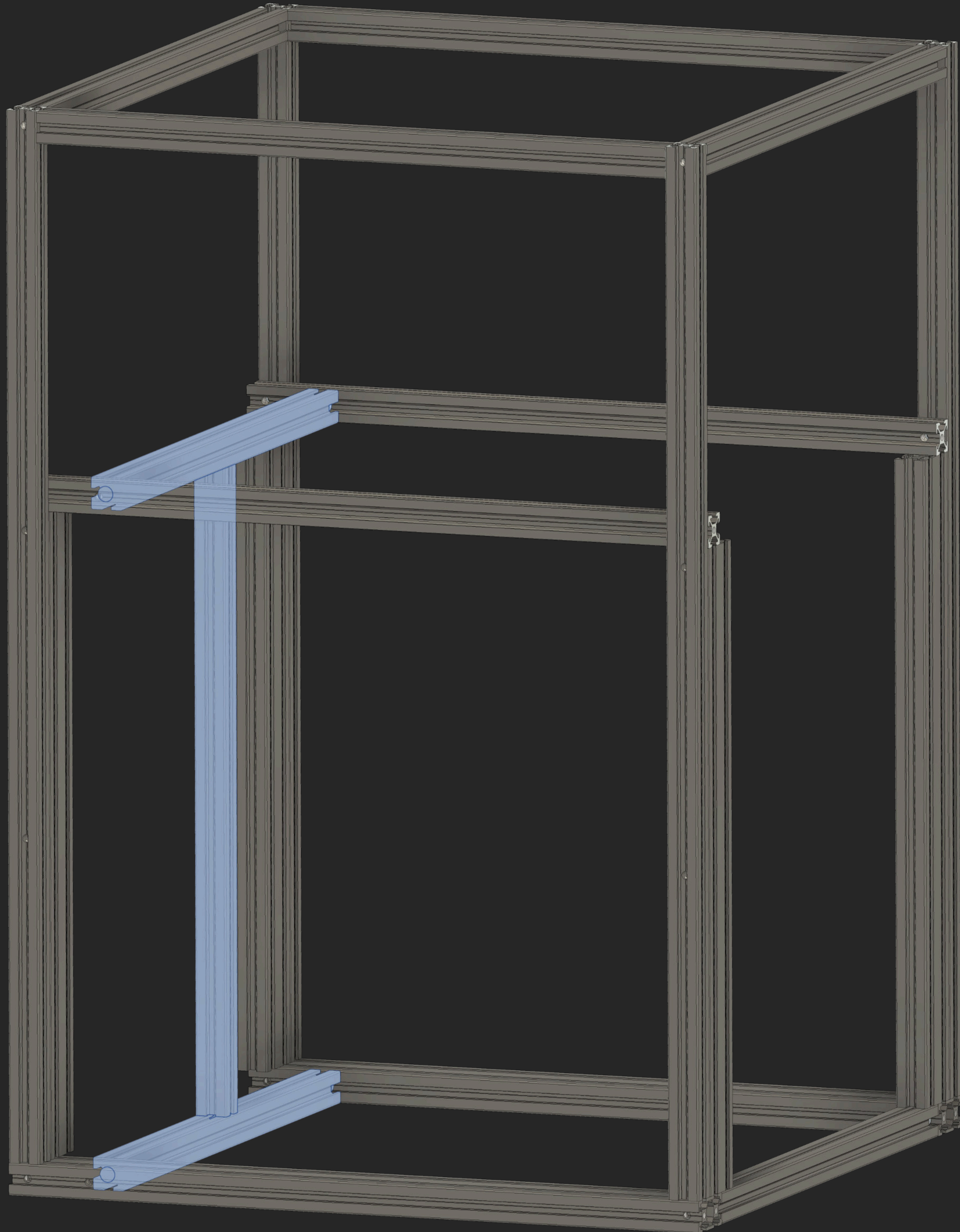
Prepare the Hydra sub-frame as shown.

	QTY.	Nebula 255	Nebula 370
G	1x	2020 400mm	2020 510mm
H	2x	2020 370mm	2020 498mm
	6x	M5 x 10 BHCS	M5 x 10 BHCS

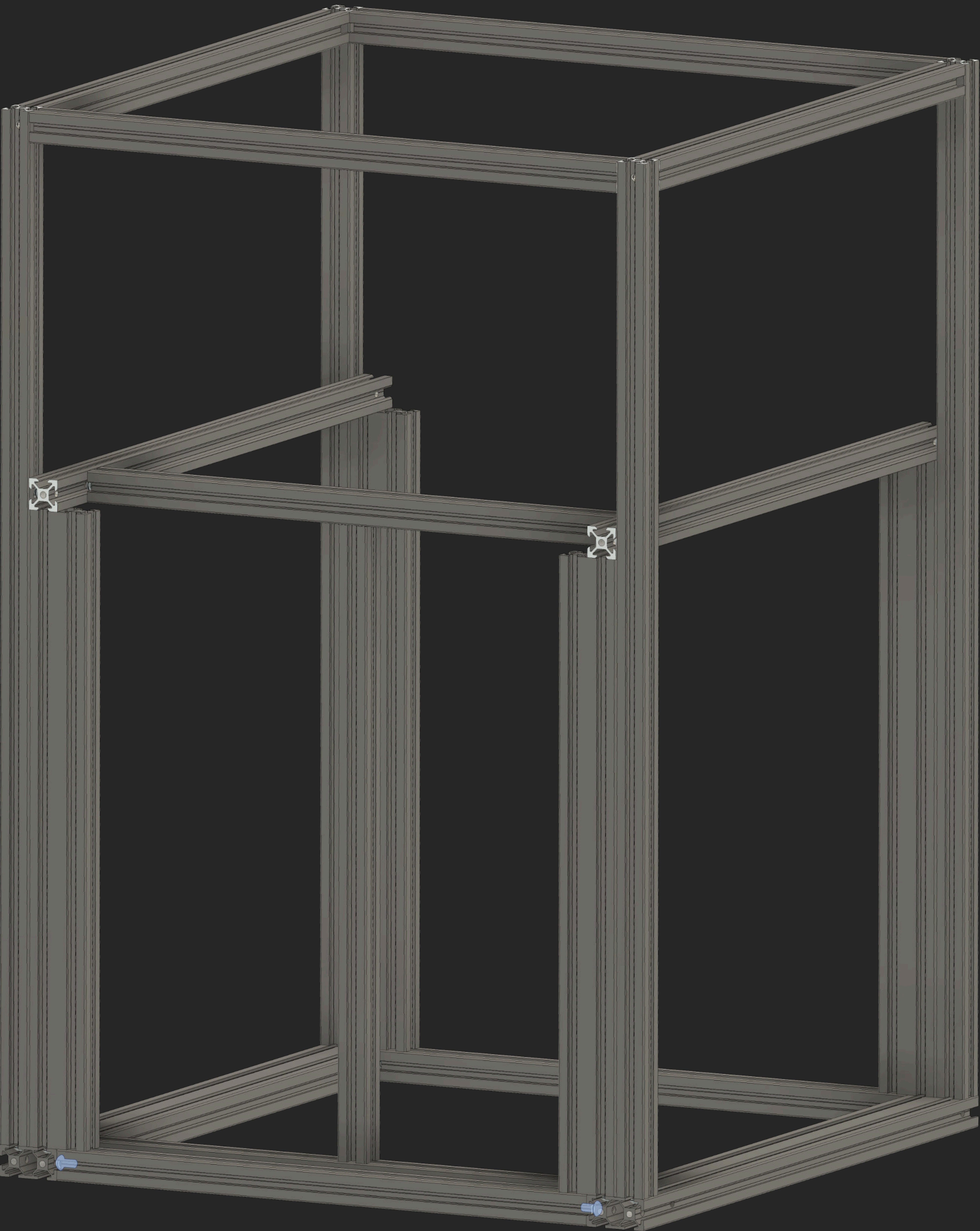


INSERT HYDRA SUB-FRAME

Slide in the Hydra sub frame, placing it 40mm from the back edge of the frame for the 255 frame, or 45mm for the 370 frame.



INSTALL FINAL REAR BOTTOM BRACE



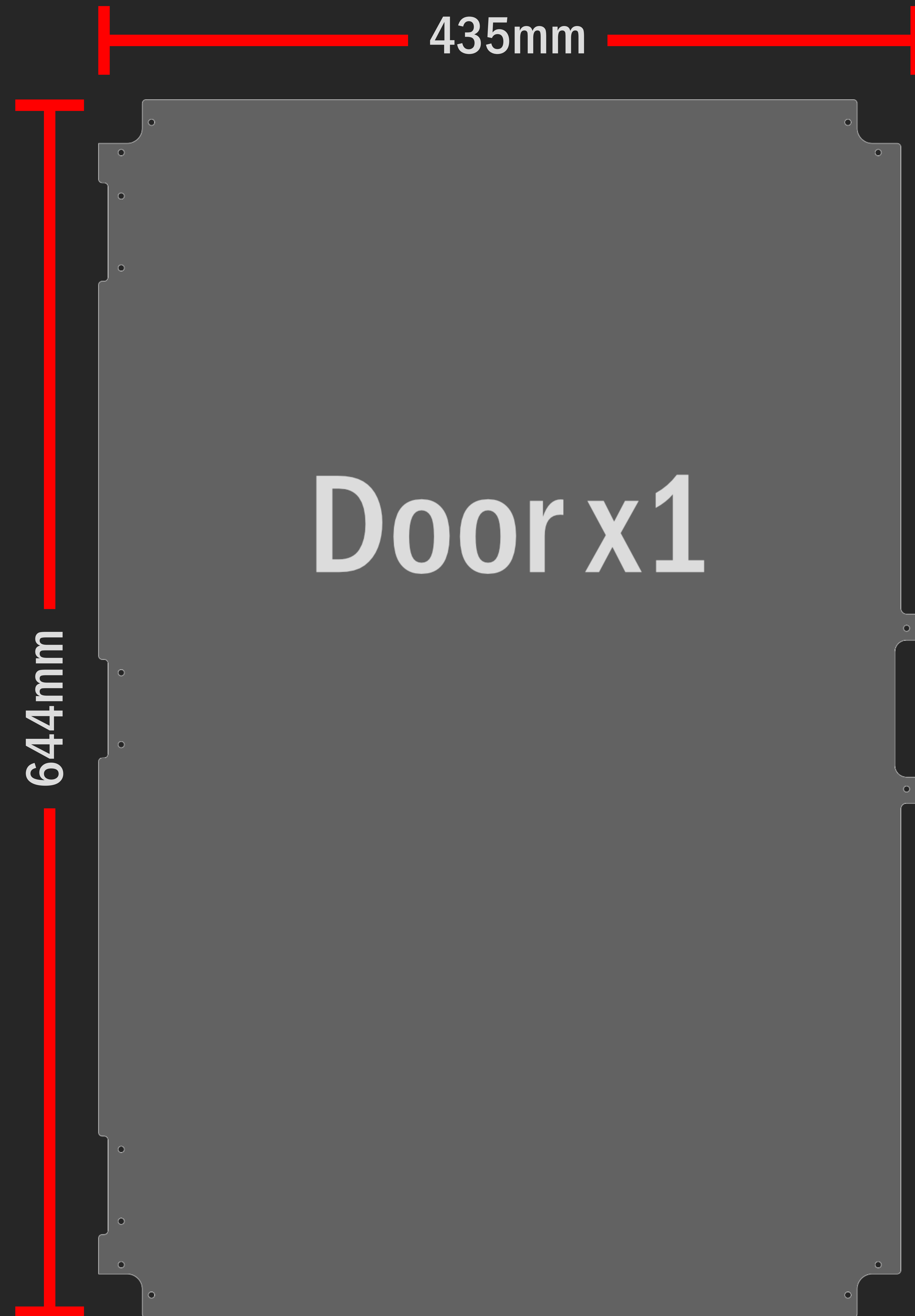
Install the final rear 2020 extrusion.

Then spend some time squaring up the frame the best you can.

	QTY.	Nebula 255	Nebula 370
I	3x	2020 370mm	2020 498m
	2x	M5 x 10 BHCS	M5 x 10 BHCS



NEBULA 255 DOOR PANEL



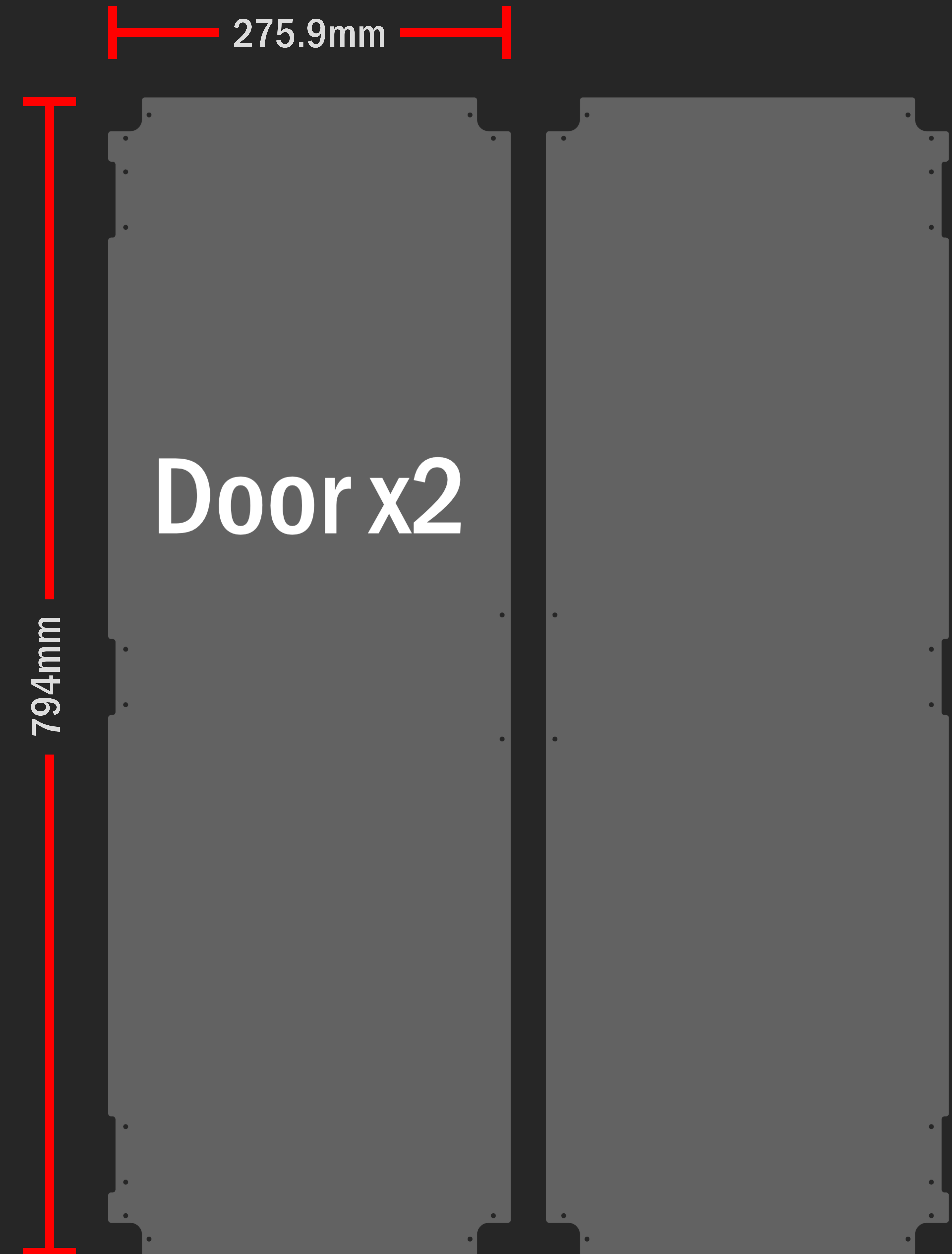
Can be laser/cnc cut.

3mm or 1/8 inch is recommended.

5.5-6mm or 1/4 inch is also supported for the door panels.

Deck Panel is the same as stock Hydra.

NEBULA 370 DOOR PANELS



Door Hinges

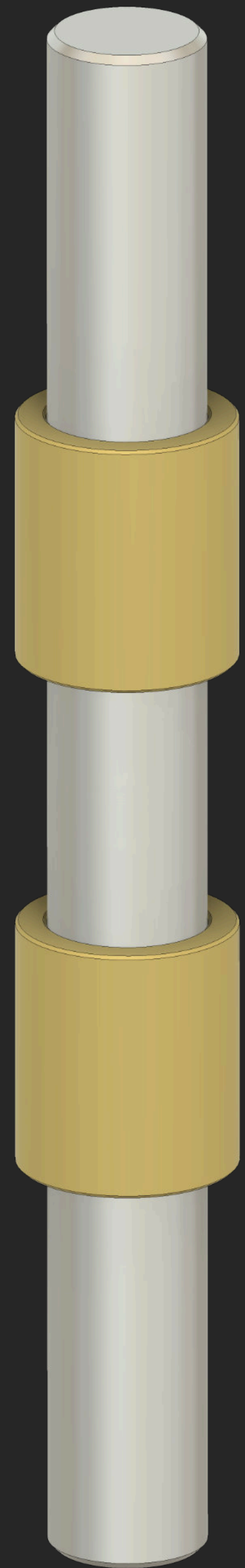


BOM per door hinge assembly
Note for Hinges, there is a ZeroG
Logo option

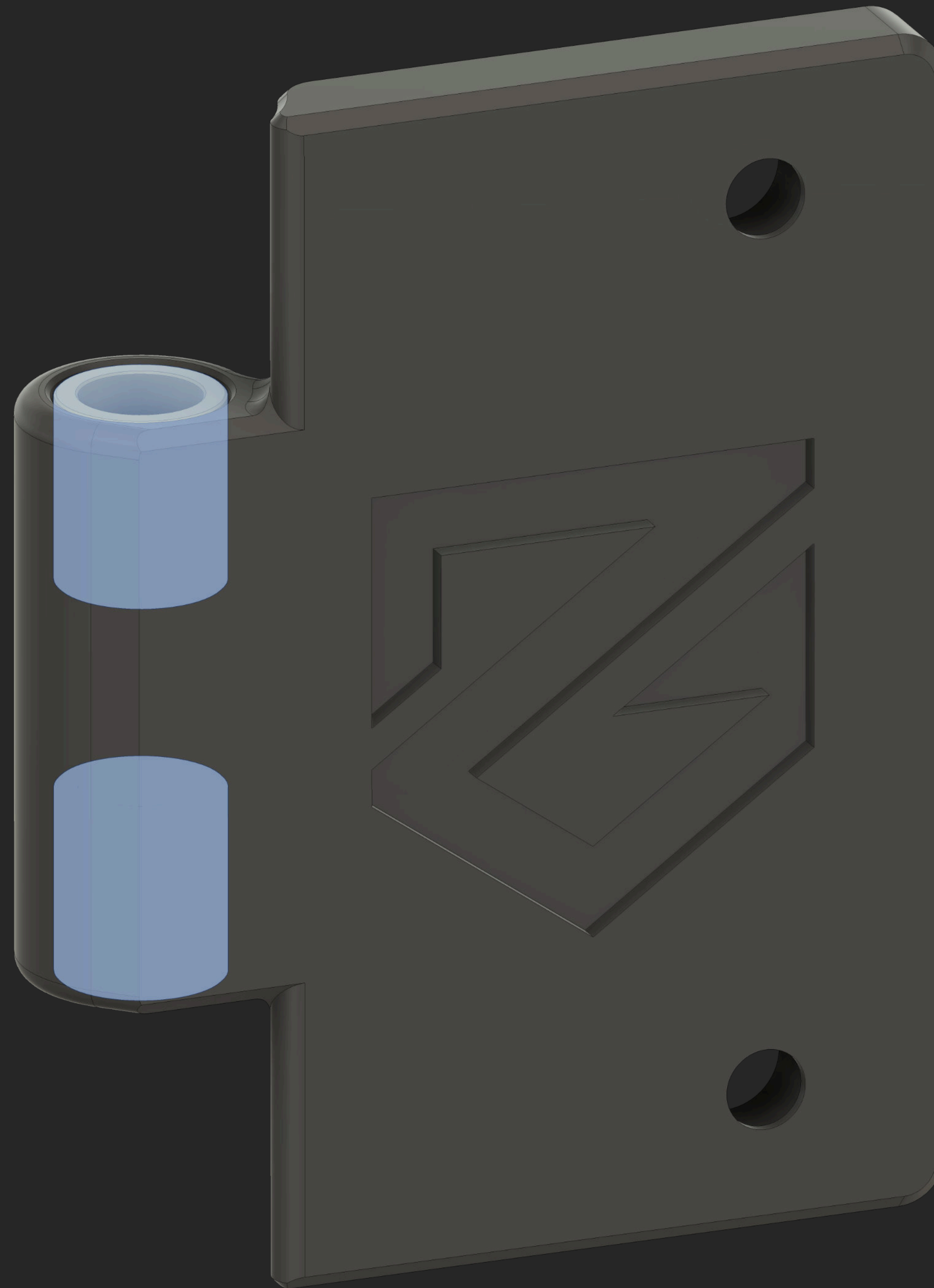
QTY.	Nebula 255	QTY.	Nebula 370
3x	Printed Base Hinge	6x	Printed Base Hinge
3x	Printed Hinge	6x	Printed Hinge
6x	5x7x8 Bronze Bushing	12x	5x7x8 Bronze Bushing
3x	5x50 Dowel Pin	6x	5x50 Dowel Pin
6x	M3x5x4 Heat-set Insert	12x	M3x5x4 Heat-set Insert
6x	M3 x 8 BHCS	12x	M3 x 8 BHCS



PIN AND BUSHINGS



Check that the bushings slide freely along the shaft. If they do not, you will need to sand the shaft.



Press the bushings in, they should be a light press fit with your fingers. If they require more force, you should adjust your print settings and reprint them or ream out the holes.



MAIN ASSEMBLY

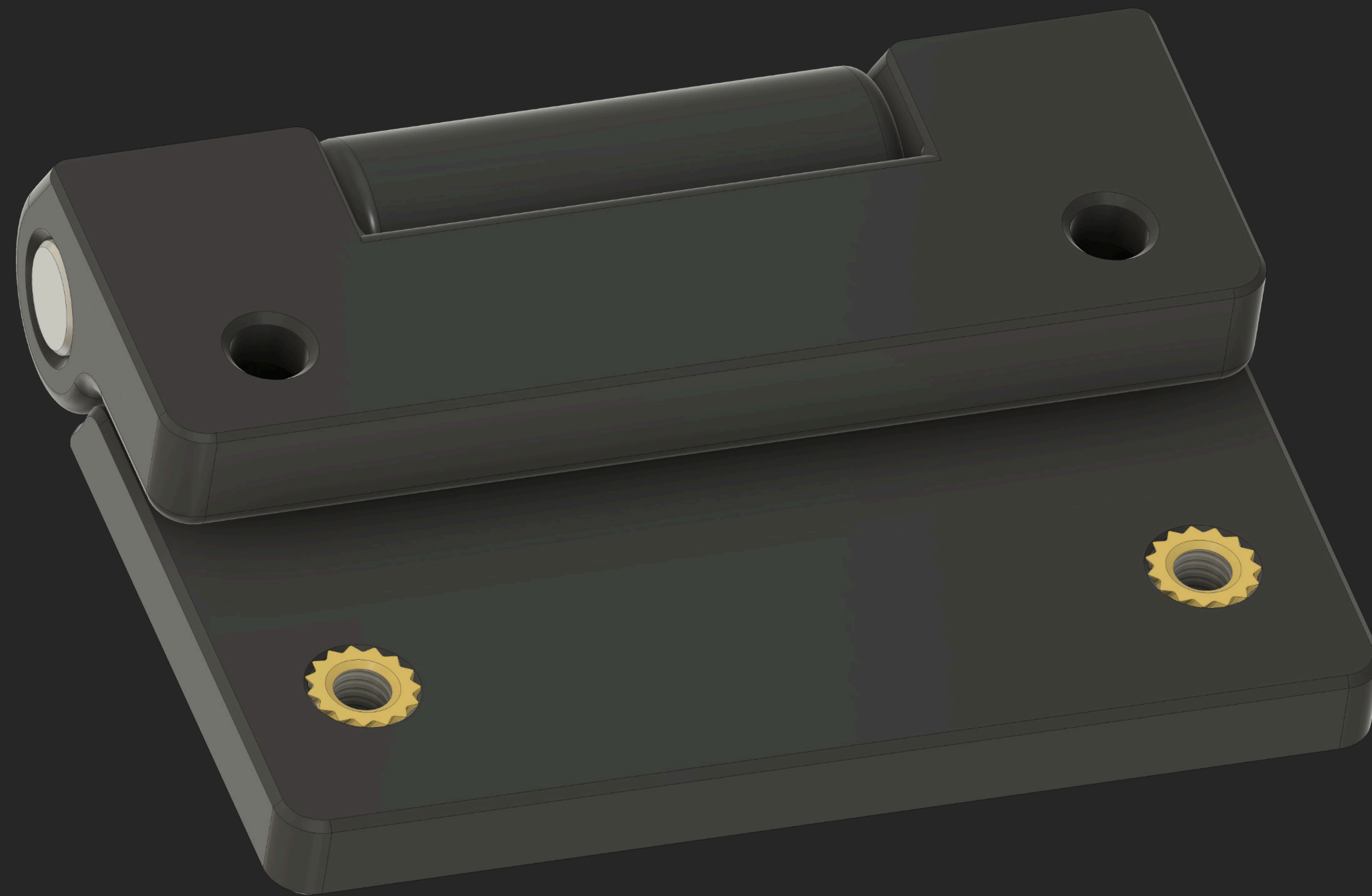


Insert the 5x50mm dowel pin that you test fit with the bushings to complete the main assembly.



INSERT HEAT-SETS

Insert 2 M3 heat-set's

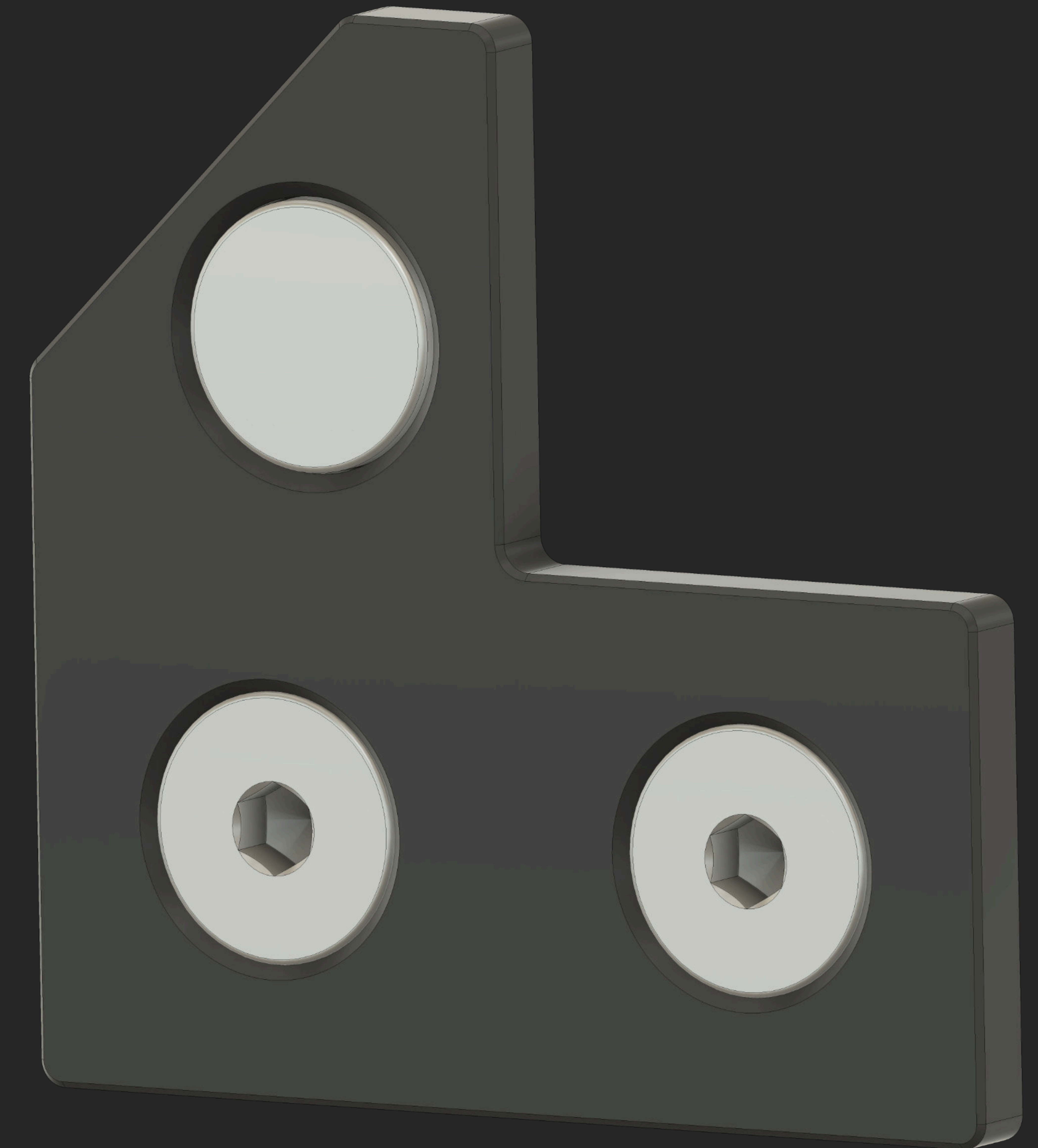
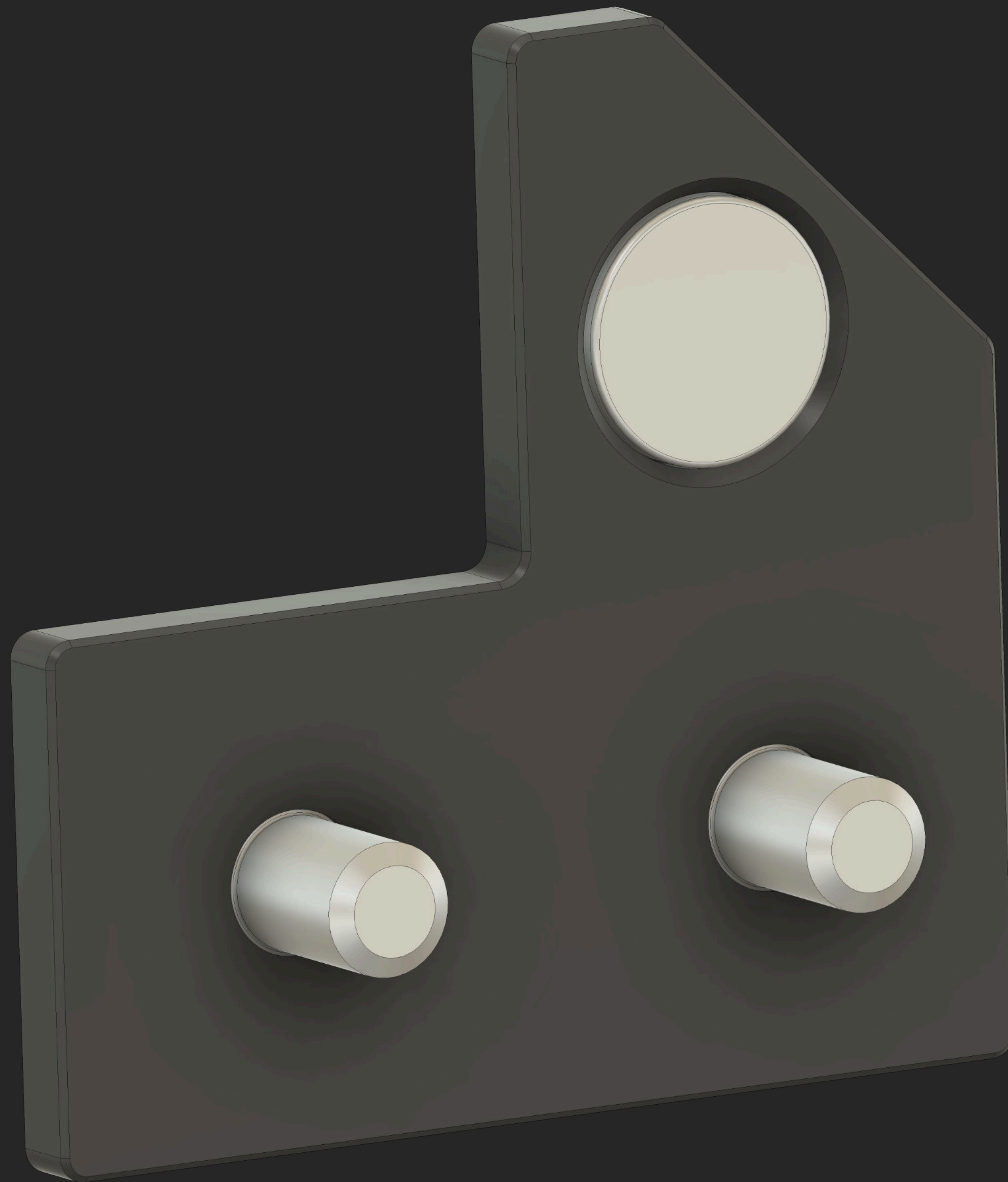


BOTTOM 2040 DOOR CATCH

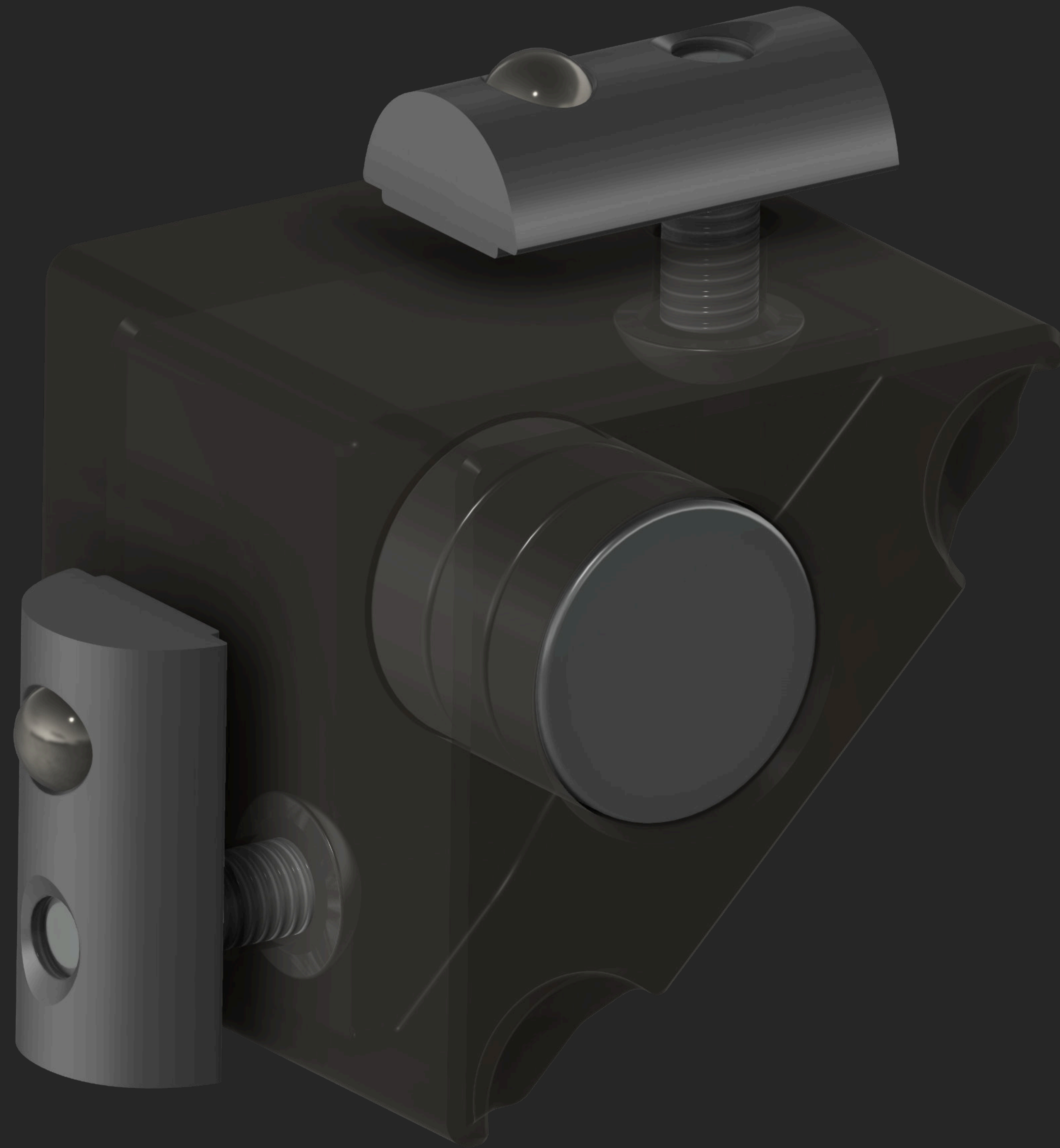
Press in a 10x3 magnet, add a couple drops of CA glue on the backside.

Install one on either side using M5x10 FHCS.

Note There is a left and right hand version.



UPPER FRAME DOOR CATCH

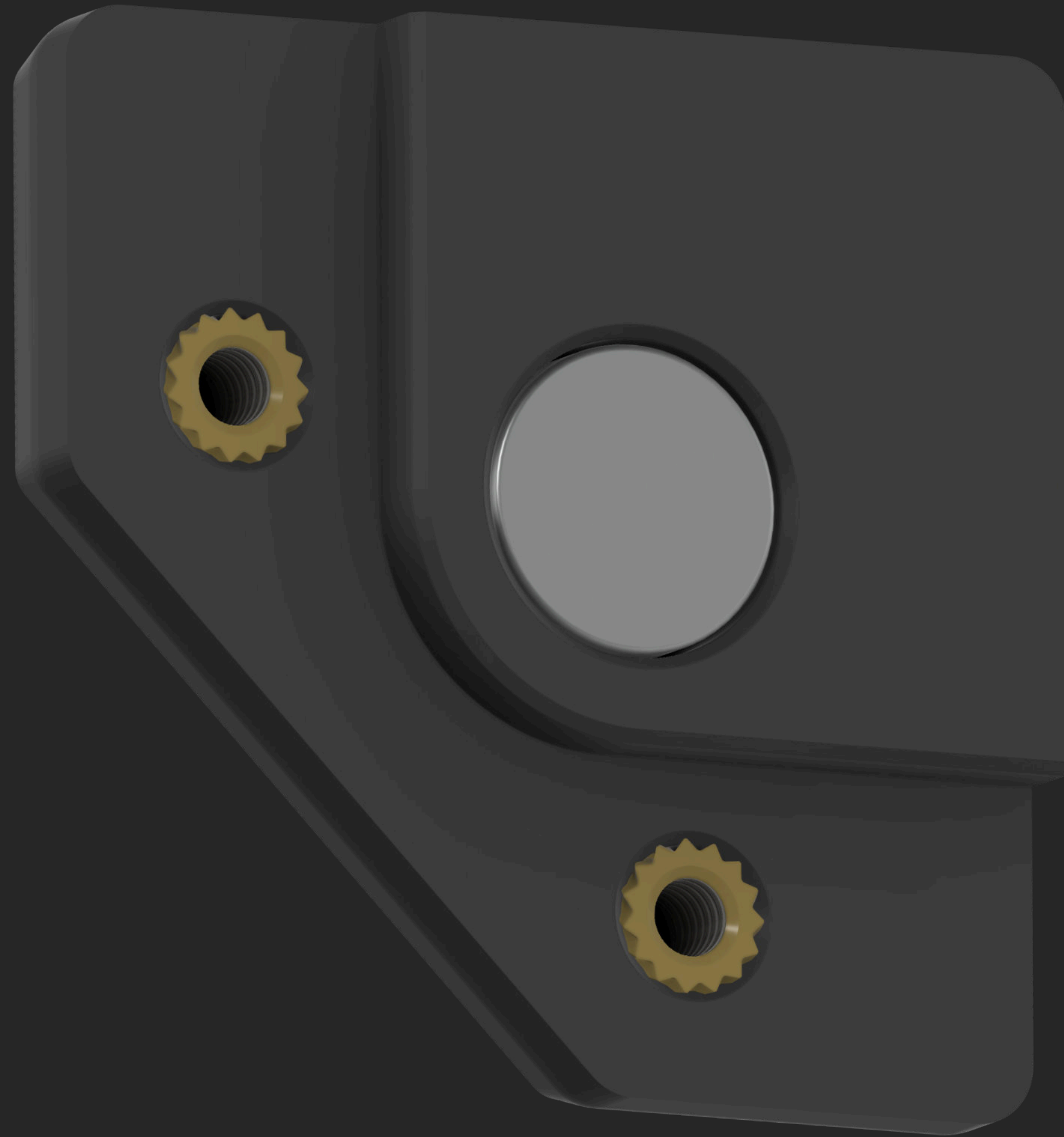


Press in 3 10x2 magnets, you can add some CA glue in the pocket before installing the magnet.

Install into both upper inside corners of the frame with the magnet facing forward using M3 T-nuts and M3x8 SHCS

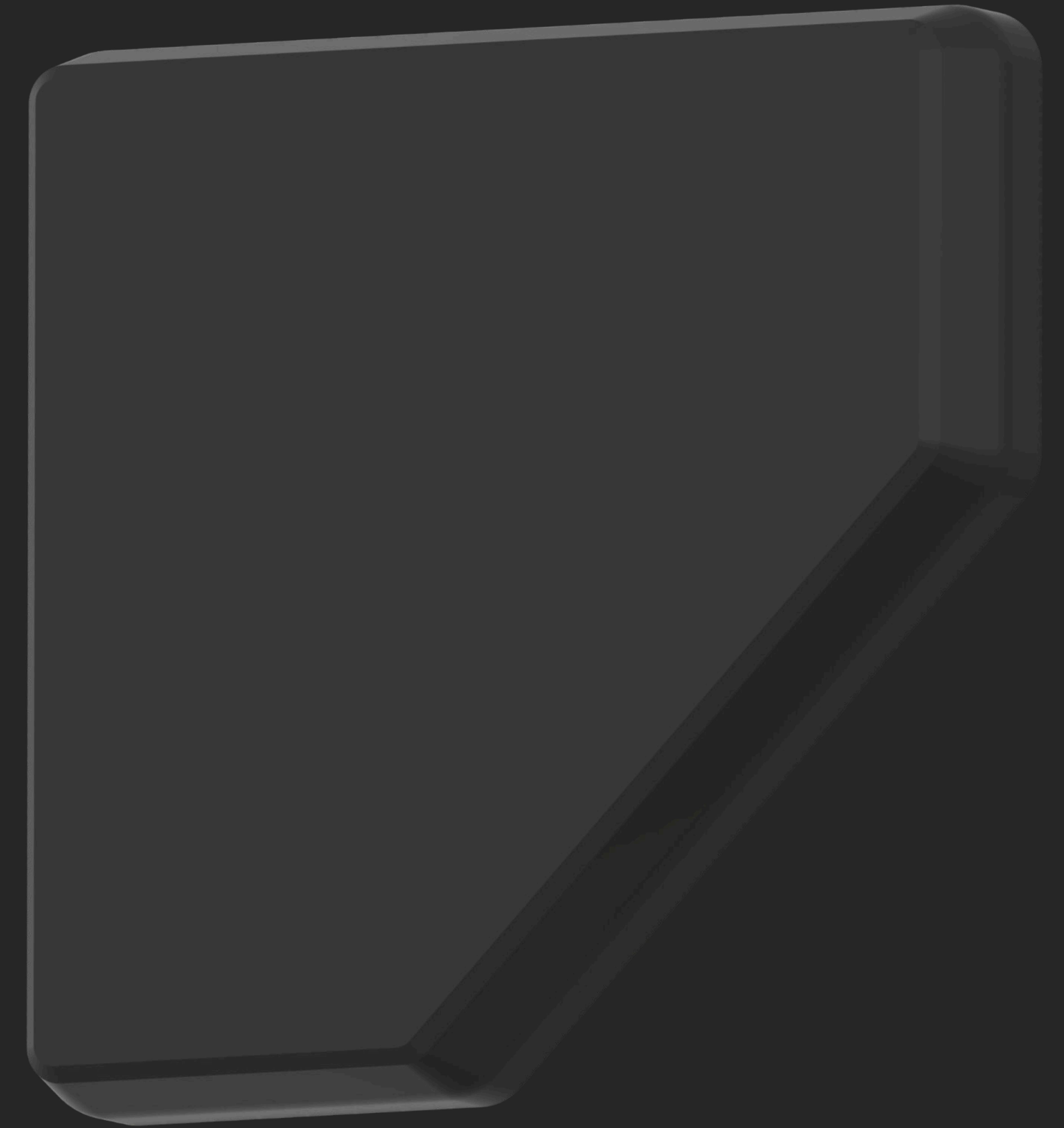


DOOR CORNER CAPS

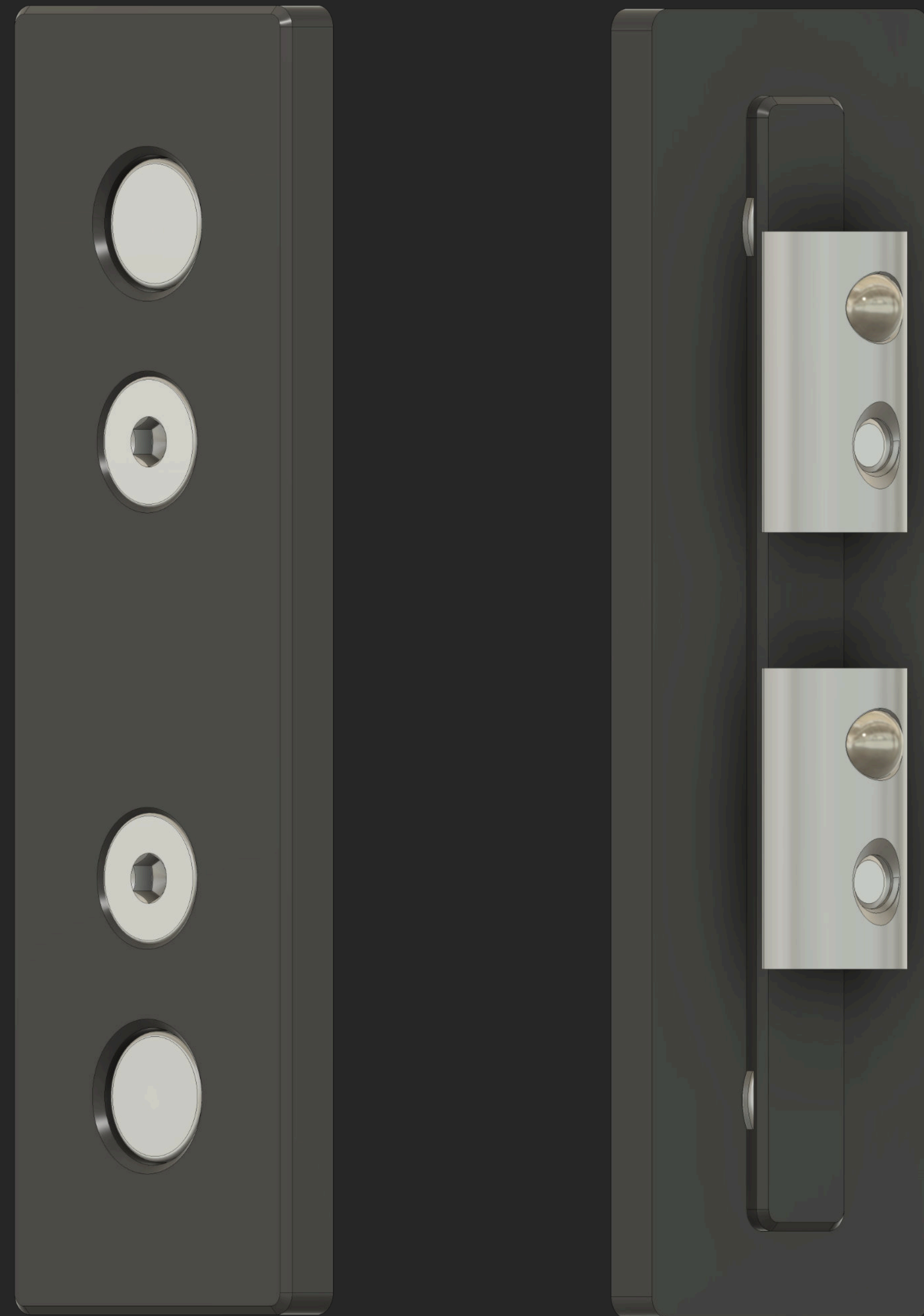


Press fit 2 10x2 magnets into the pocket. Install heatsets in the other holes.

Install one in each corner using M3x8 BHCS



255 MAGNETIC DOOR CATCH



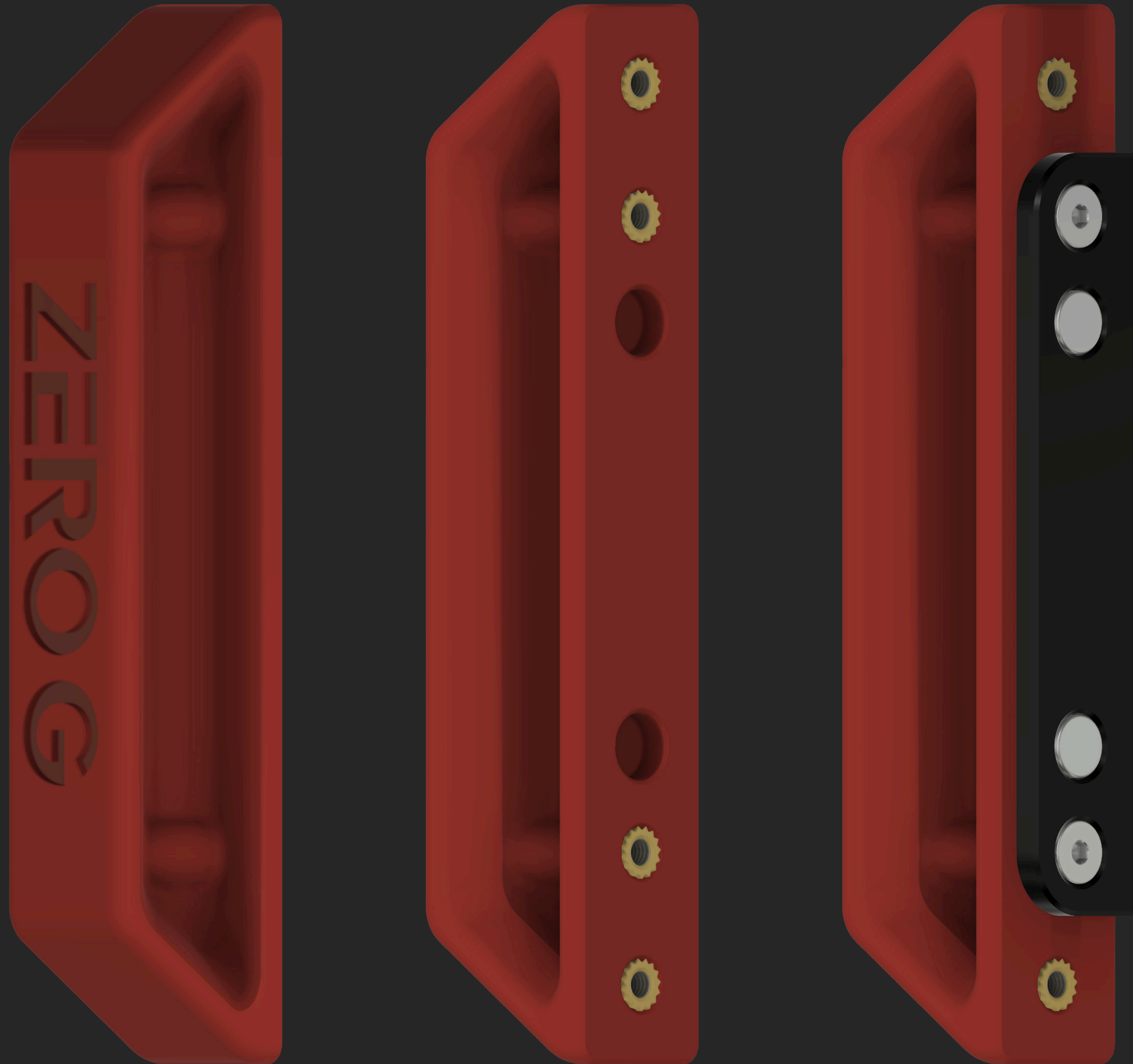
Apply some CA glue into each magnet pocket, then press fit the 6x3mm magnets.

Install onto the 2020 upright using M3x8 FHCS and M3 T-nuts.



255 Door Handle

Install 4 M3 heat-set inserts into the main handle.
Attach the spacer to the handle using 2 M3x8 FHCS.
Glue 2 6x3 magnets into each pocket. Then install
onto the door using M3x8 BHCS



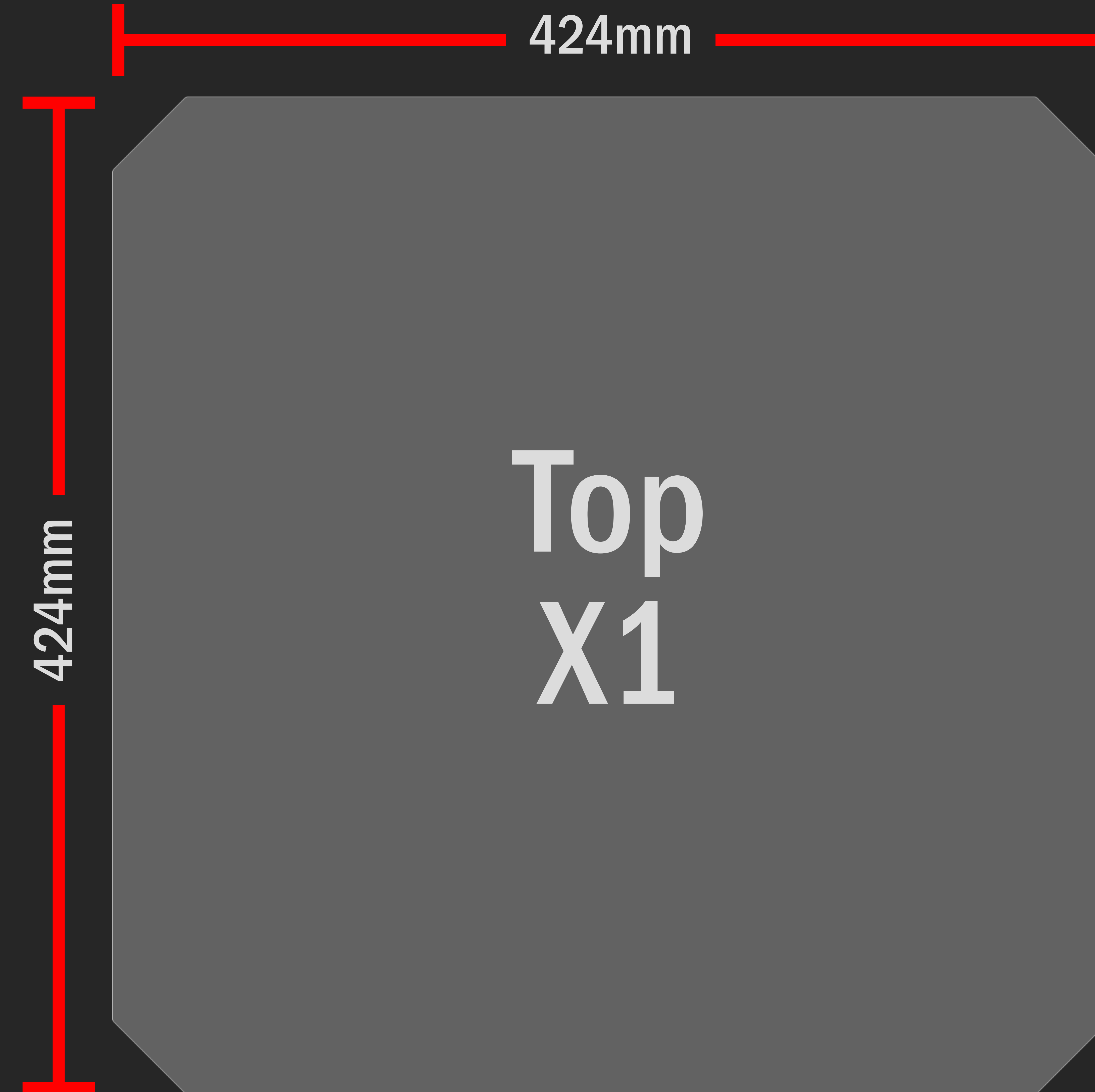
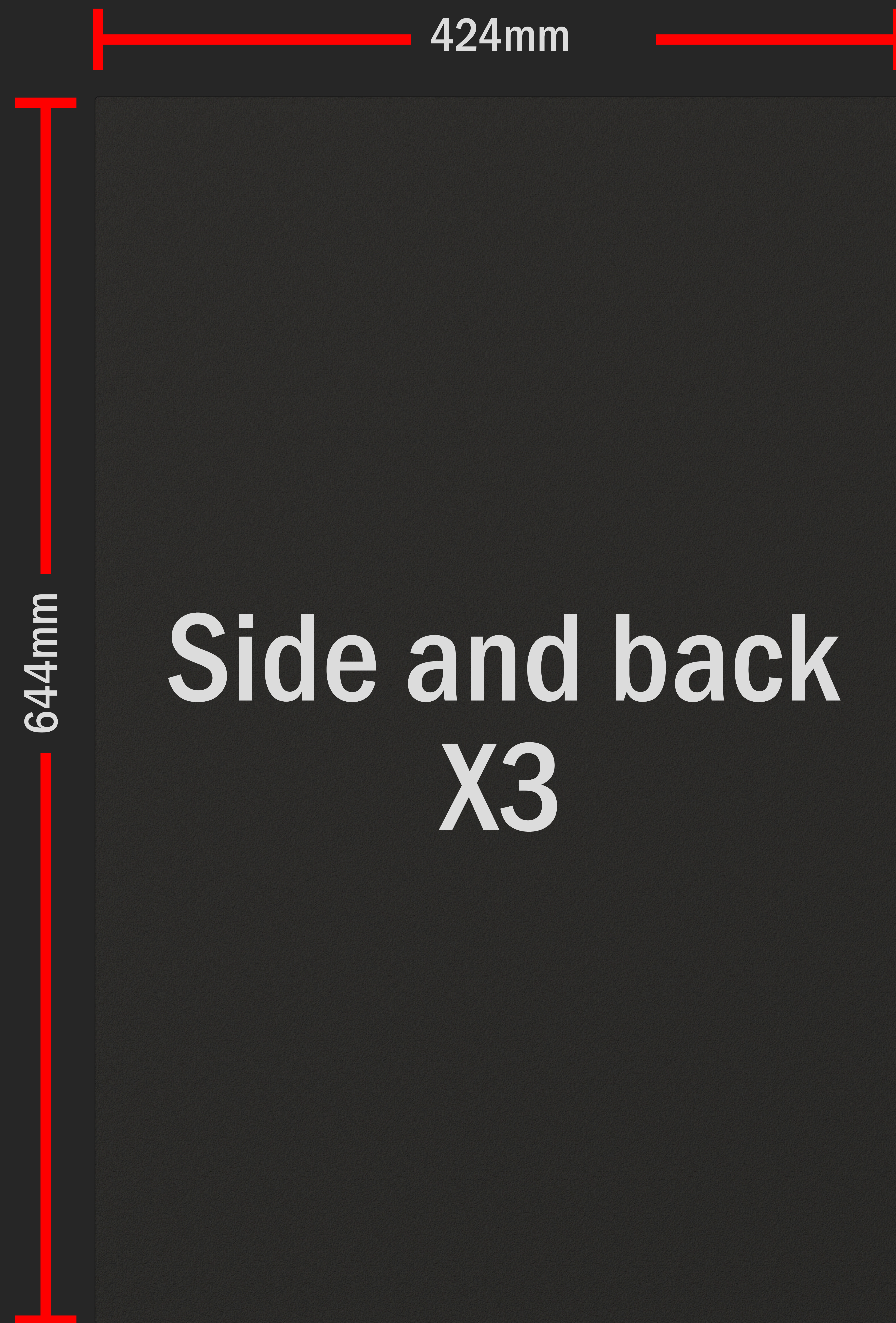
370 Door Handle



Install two M3 heat-set inserts into each handle, then press in two 10x2mm magnets into each magnet pocket. Attach to the door using M3x8 BHCS.



NEBULA 255 SIDE AND TOP PANELS



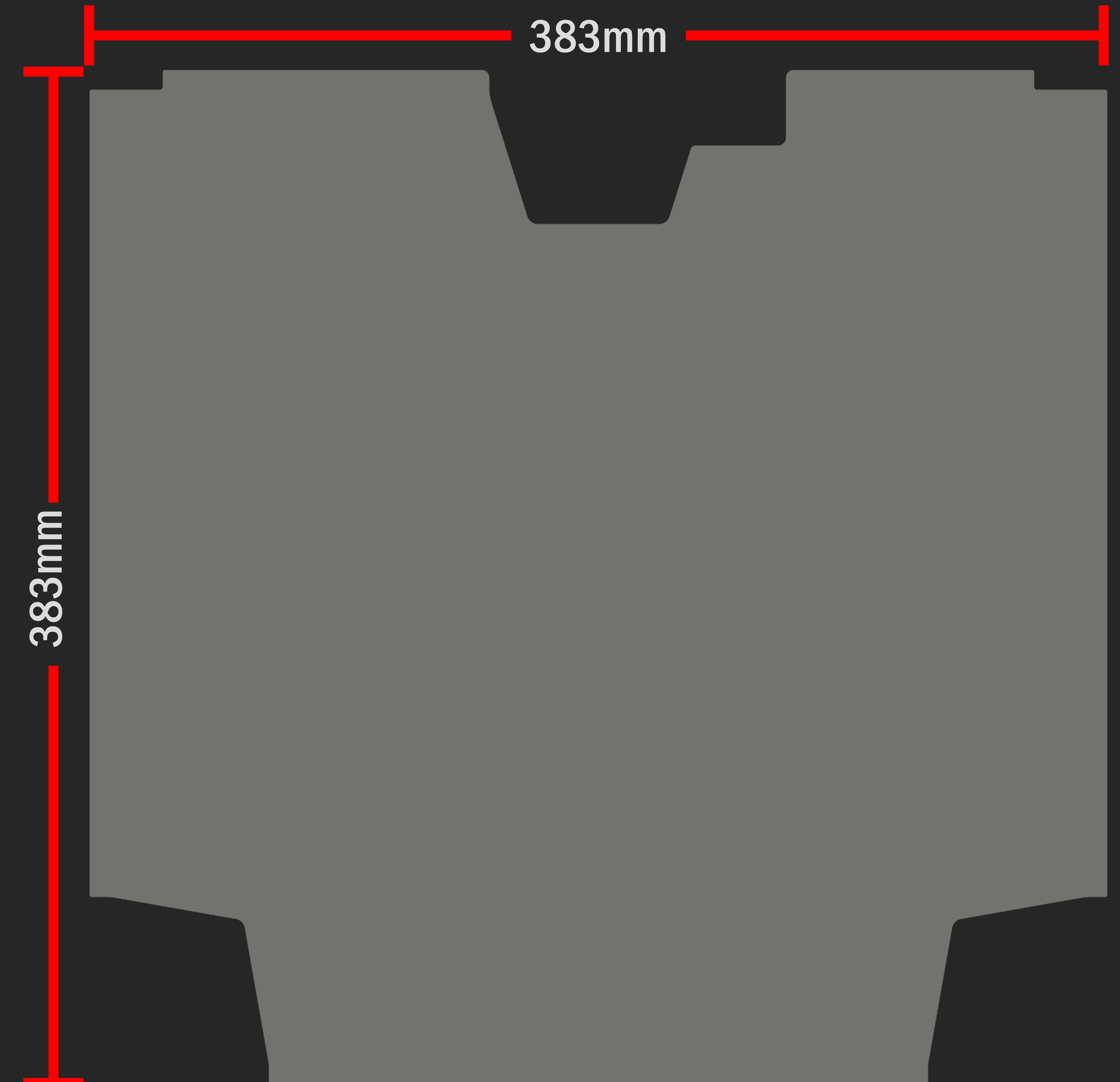
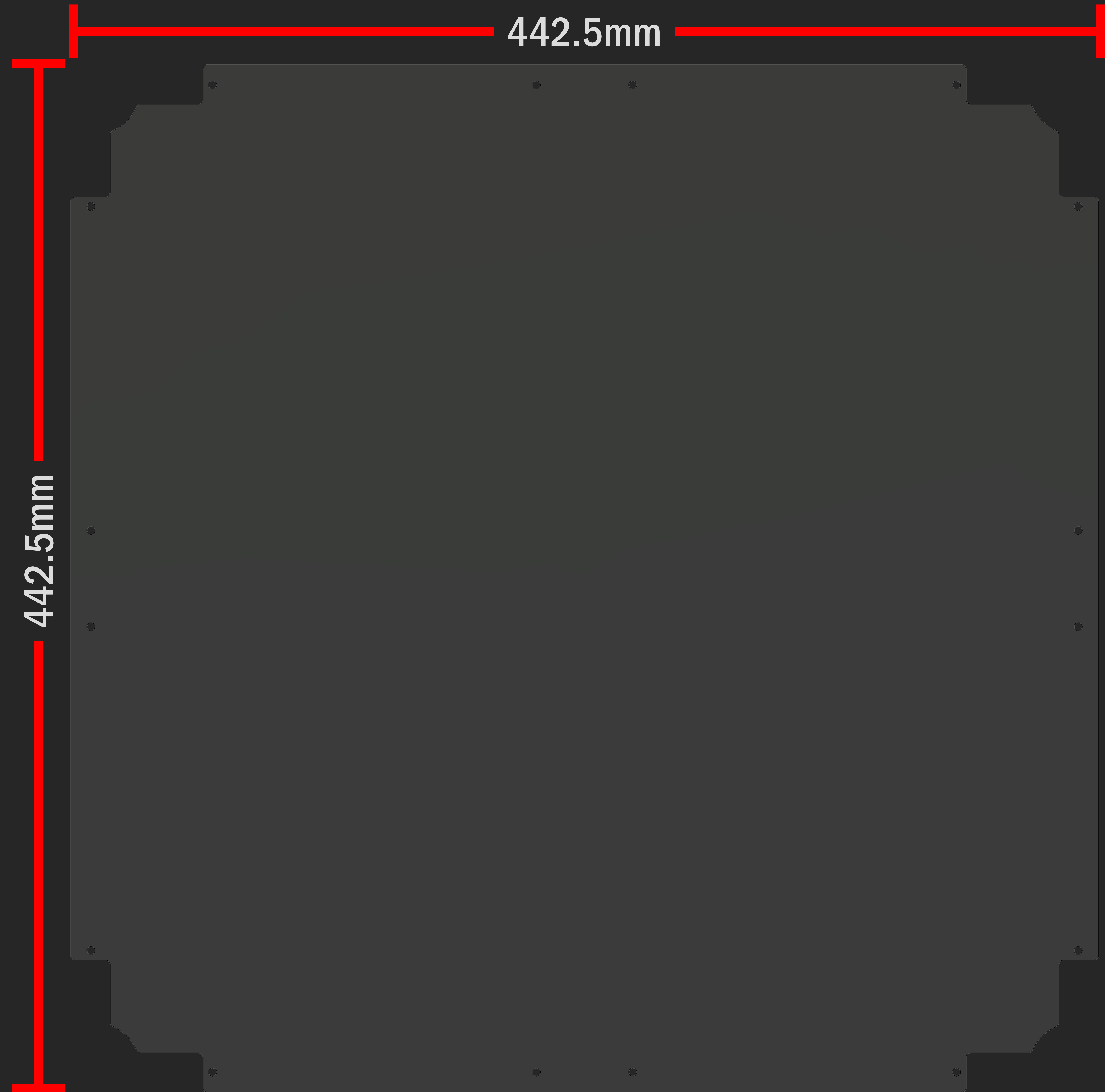
Can be saw cut, or laser/cnc cut.

3mm or 1/8 inch is recommended.



NEBULA 255 ELECTRONICS BAY PANELS

- Laser or CNC cut
- 3mm or 1/8 inch thick
 - Top panel as same as stock Hydra panel



NEBULA 370 SIDE, BACK, AND TOP PANELS

Side Panel X2

574mm

792mm

Back Panel X1

552mm

792mm

Top Panel X1

542mm

574mm

Can be saw cut, or laser/cnc cut.

3mm or 1/8 inch is recommended.



NEBULA 370 ELECTRONICS BAY PANELS

- Laser or CNC cut
- 3mm or 1/8 inch thick
 - Top panel as same as stock Hydra panel

Bottom Extension

570.5mm

Bottom Panel

547.5mm

511mm

Deck Panel

528mm

